Lesson ①:

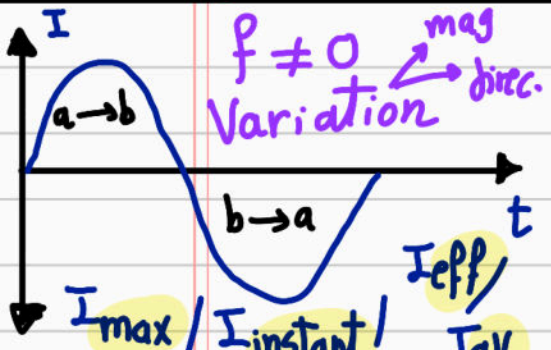
Egypt (220V - 50Hz)

Chapter ④
AC Circuits

Content:
1. Ac Vs Dc ☑
2. Hot Wire Ammeter ☑
3. R-Circuit ☑
4. L-Circuit ☒
5. Capacitor ☑
6. C-Circuit ☐

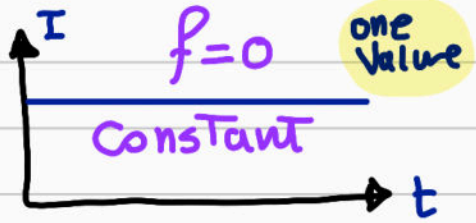
1 Ac Vs Dc:

Alternating Current



$f \neq 0$ Variation ↔ mag, direc.

Direct Current



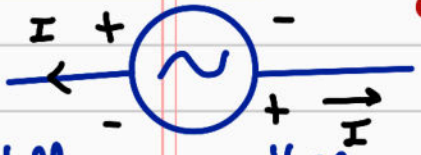
$f = 0$ one Value

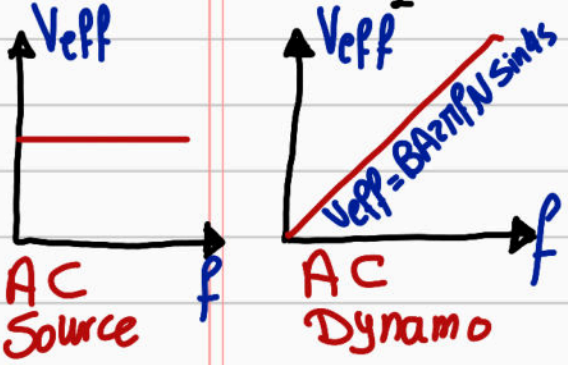
Constant

Constant ↔ magnitude, direction

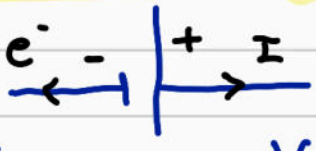
No average
No effective
No max
No instant

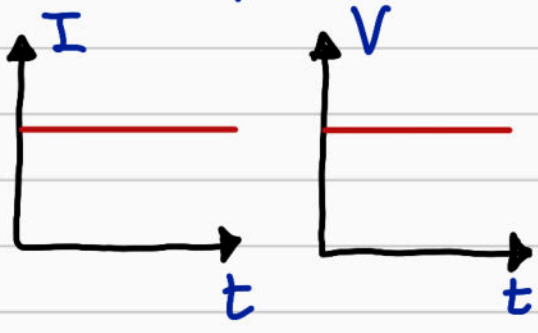
AC Source or AC Dynamo





DC Source or Battery





1. AC → (Rectification) → Dc
2. Step up or down (Transformer)
3. Thermal effect (effective Valve) $P_w \propto w$

1. Dc → (✗) → Ac
2. Cannot be stepped up or down (No Variation No Induction)
3. Thermal effect ↓↓



1. Power Transmission
2. Lighting System
   1. Home 2. factory
   3. Street 4. Work

Physics

1. Electroylsis
2. Electroplating
3. Charging Batteries



chemistry*

[2] Hot Wire "Thermal" Ammeter:

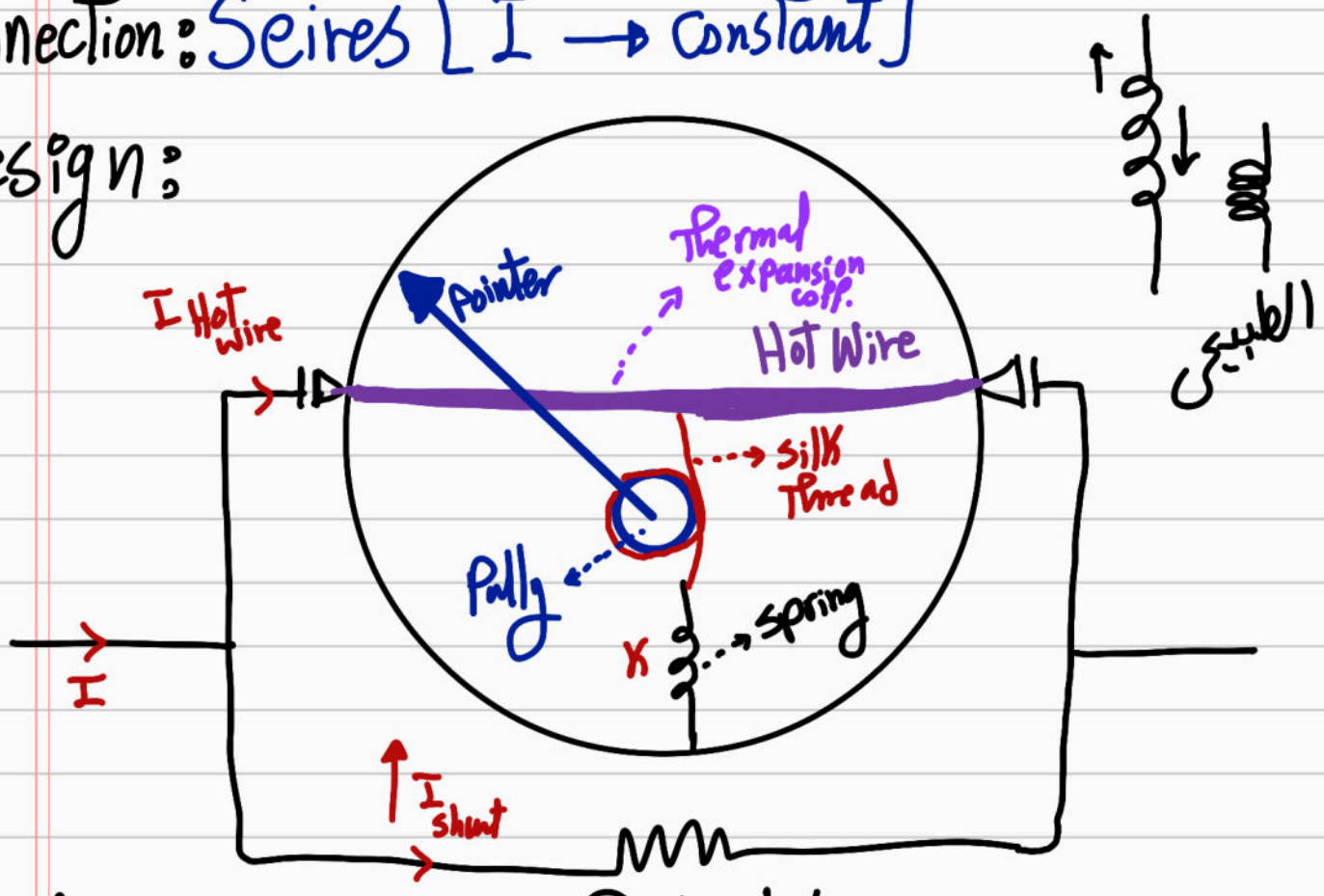
AC ↗ DC → Any Type (I)

#Sci. idea: thermal effect of electric Current

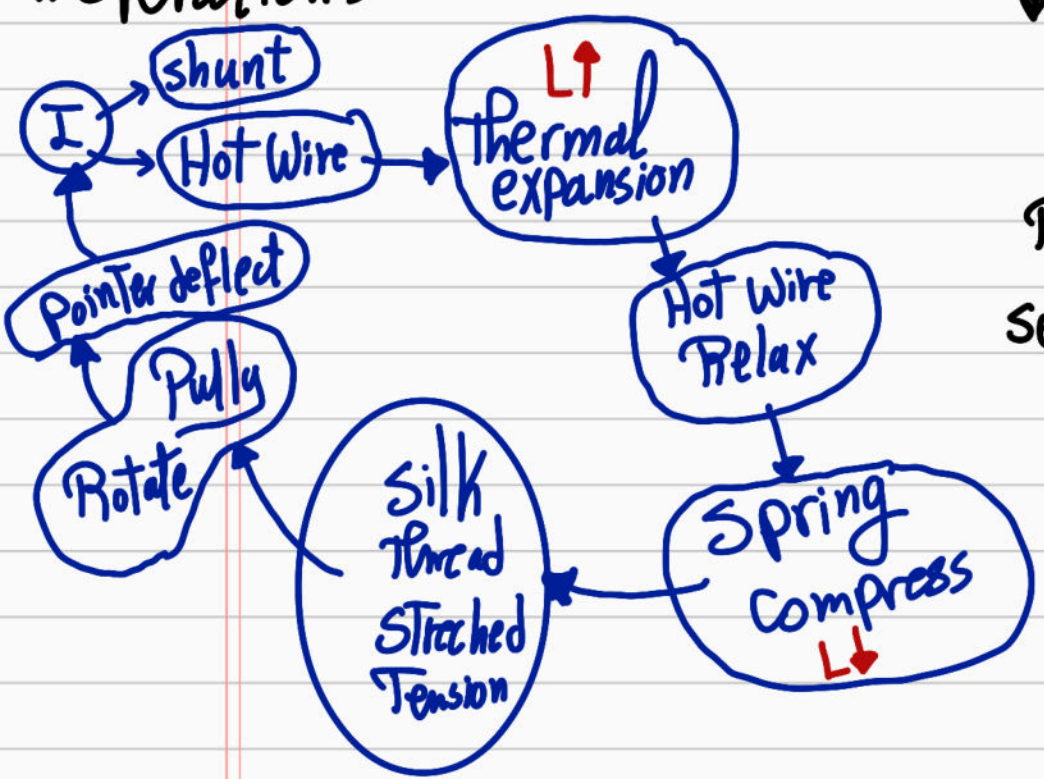
#uses: 1. measure Dc 2. measure effective Value Ac

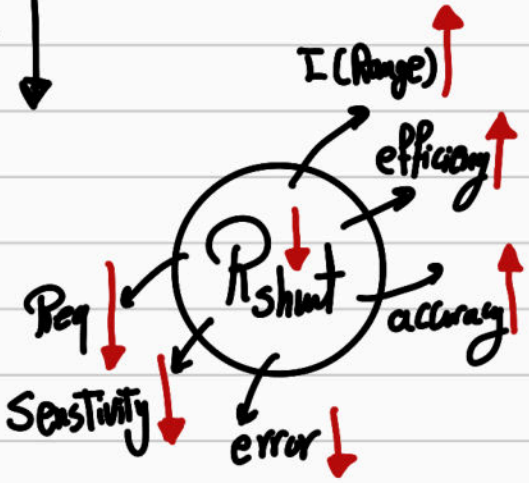
#Connection: Seires [I → constant]

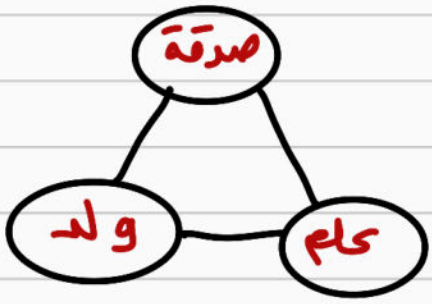
#design:



#operation:

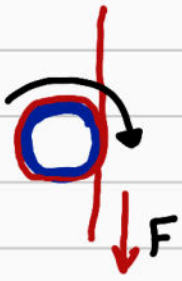






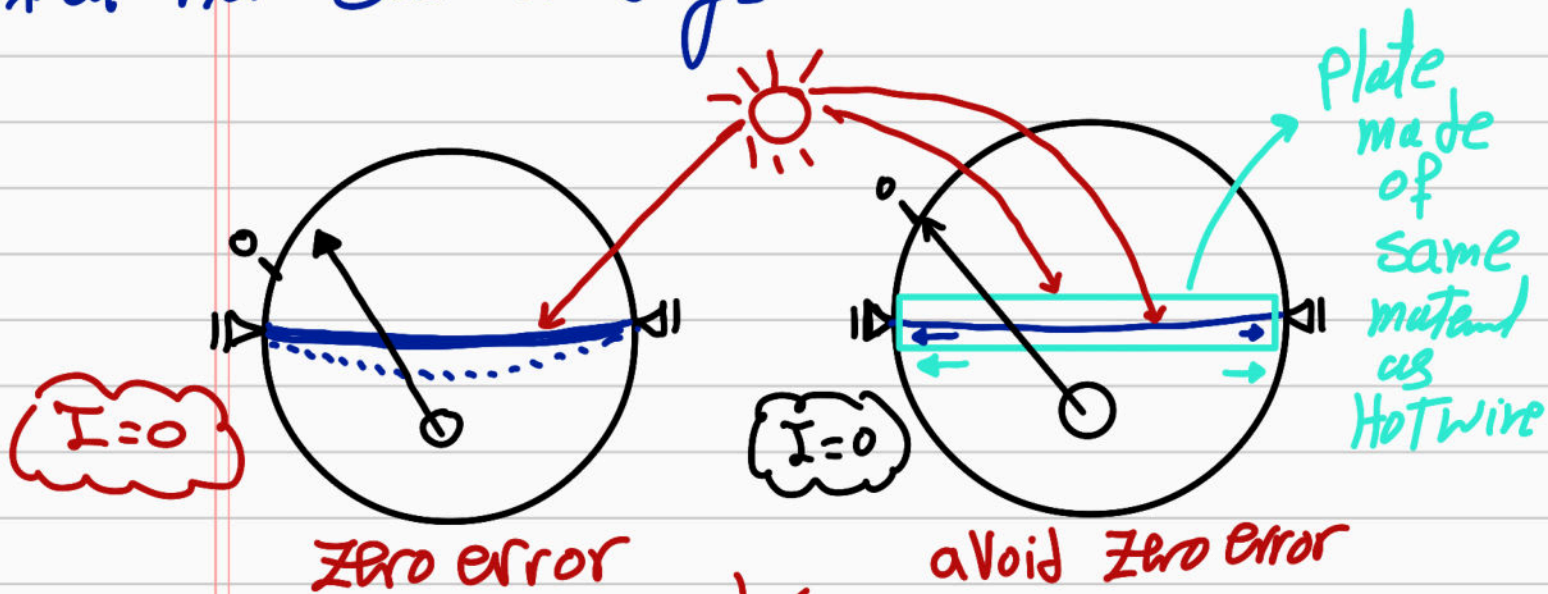
#Components:

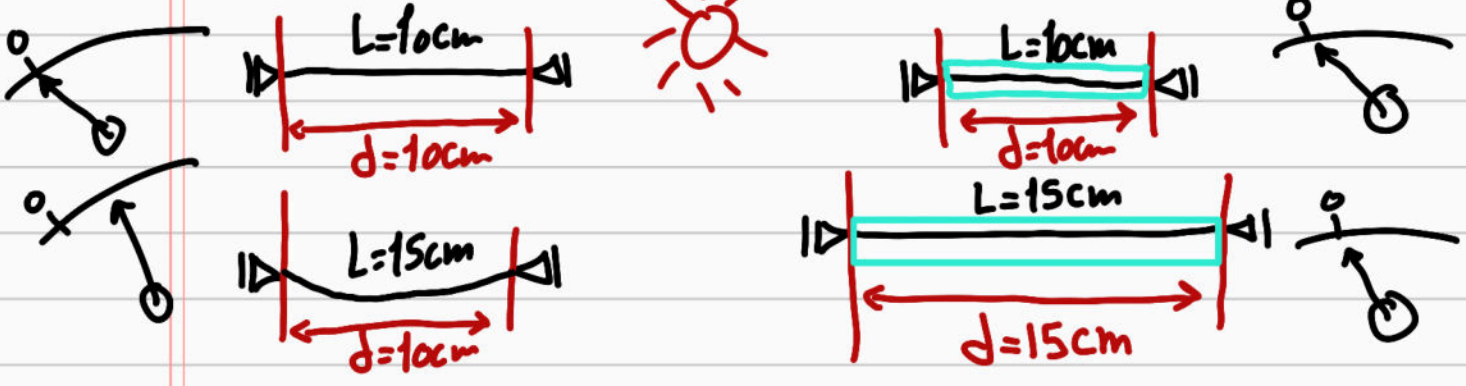
* spring → Stretch / Tension Silk Thread
* Pully → Rotate To deflect pointer
* silk Thread → Connect ← spring, Pully, Hot Wire

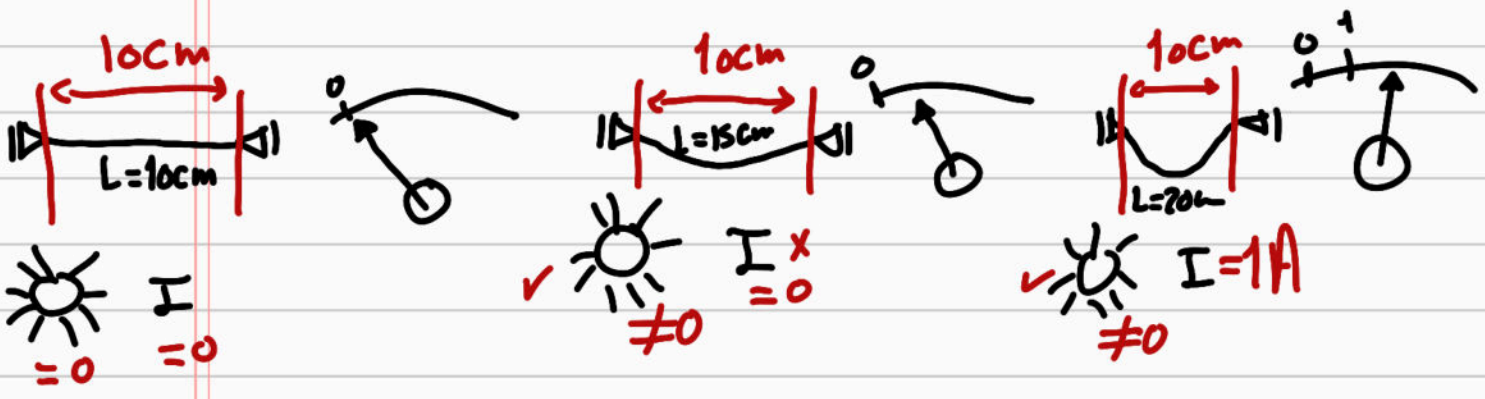


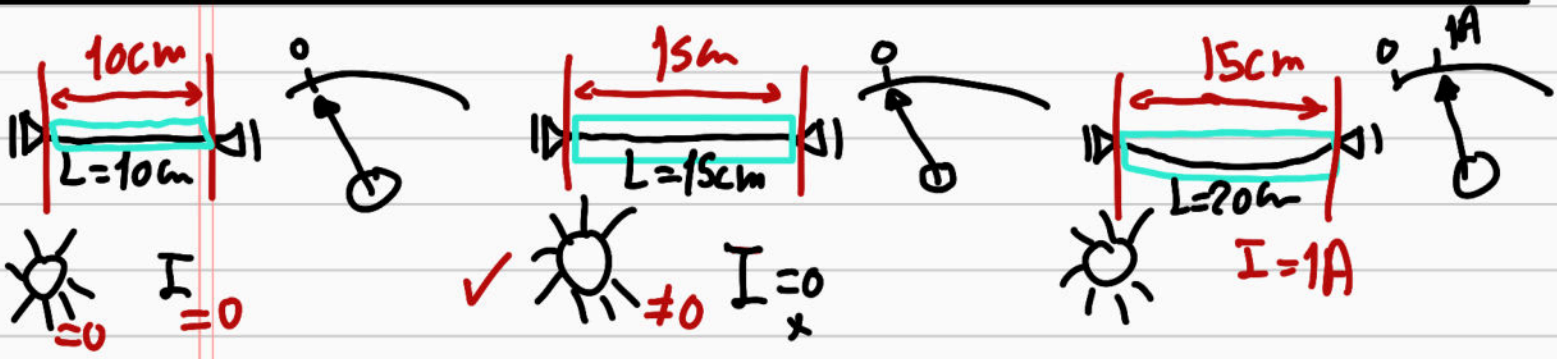
* Hot Wire → Platinume & Irradiume alloy
  1. High Melting Point
  2. High Thermal expansion coff.
  3. Heat Senstivie* → Zero error

* at hot Summer days







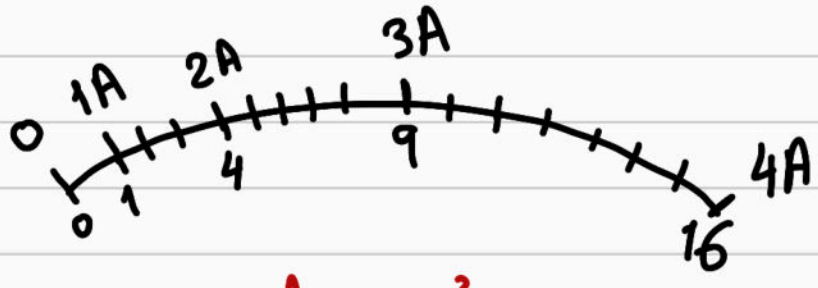


#Scale is non-uniform

thermal energy $= P_w \times t$

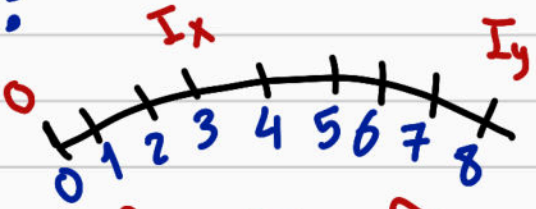
$= I_{eff}^2 R t$

$= \frac{V_{eff}^2}{R} t$



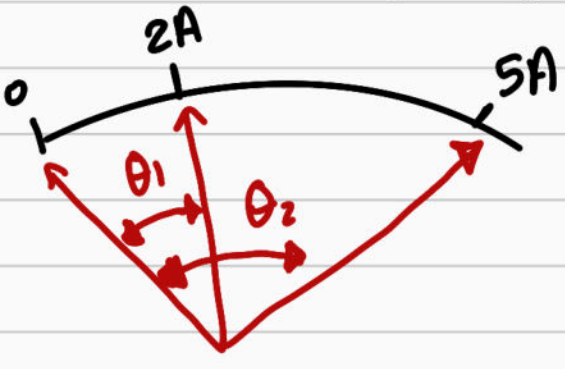
$\text{Scale} \propto I_{eff}^2$

examples:



$\frac{I_x}{I_y} = \sqrt{\frac{3}{8}}$

$\theta \propto \text{scale} \propto I^2 \rightarrow I \propto \sqrt{\text{scale}}$



$\frac{\theta_1}{\theta_2} = \left(\frac{I_1}{I_2}\right)^2 = \left(\frac{2}{5}\right)^2$

$= \frac{4}{25}$

thermal expansion coefficient is α



$\alpha_{plate} < \alpha_{wire}$

the reading is above normal

Greater than θ



$\alpha_{plate} = \alpha_{wire}$

The reading is normal

"avoid zero error"

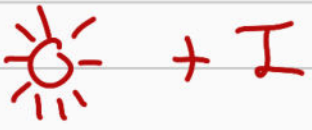
θ



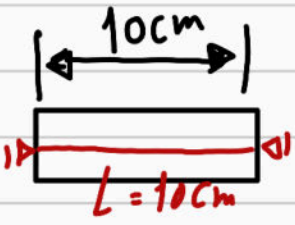
$\alpha_{plate} > \alpha_{wire}$

The reading is below normal

less than θ

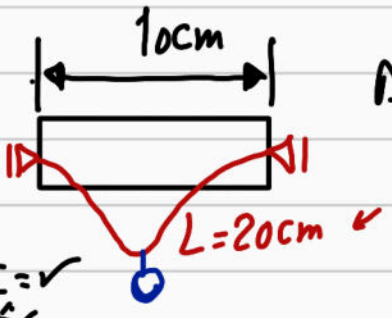


$+I$



$I=0$

$=0$



$I=r$

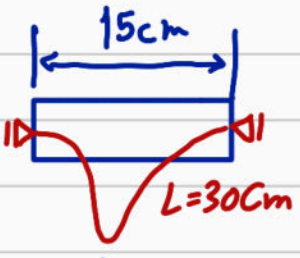
$=0$

$\Delta L = 10cm$

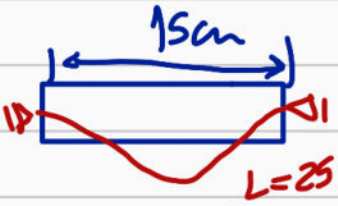
$\alpha_{wire} > \alpha_{plate}$ Reading $> \theta$

$\alpha_{wire} = \alpha_{plate}$ Reading $= \theta$

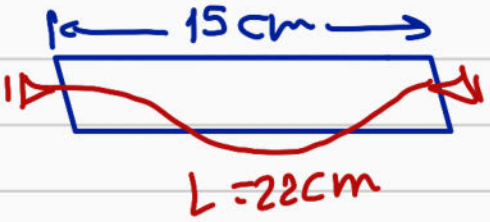
$\alpha_{wire} < \alpha_{plate}$ Reading $< \theta$



$\Delta L = 15cm$



$\Delta L = 10cm$

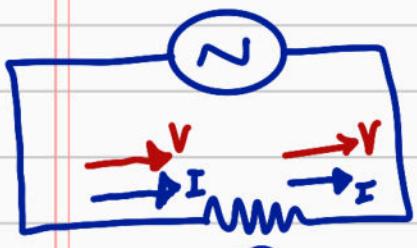


$\Delta L = 7cm$

3 non-Inductive ohmic Resistance [R-Circuit]

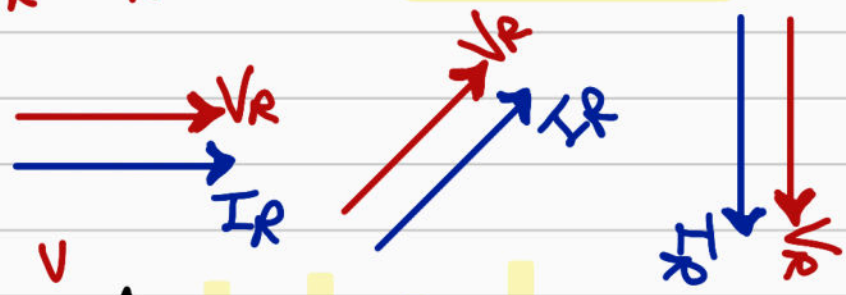
"L = 0" self induction coff.

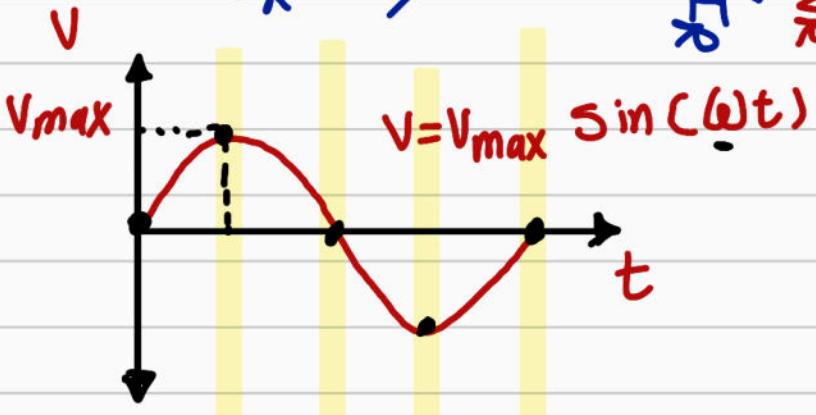
#Circuit:

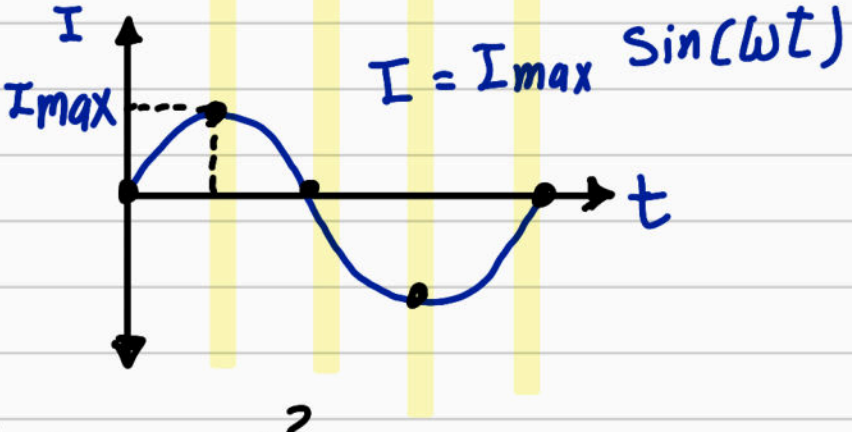


Consume R
not delay

V_R & I_R in the same phase **







#Rules

① $V = I R$

② $R = \frac{\rho L}{A}$

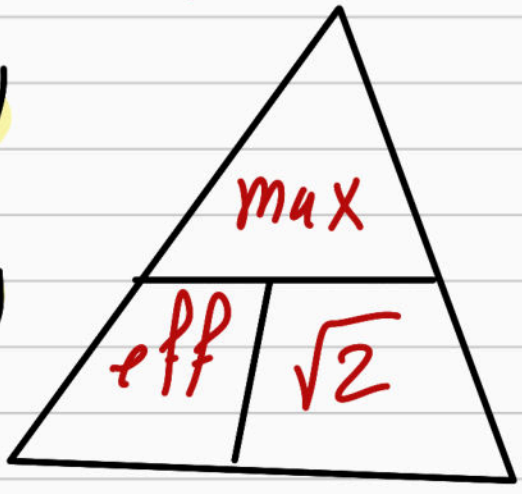
③ $\rho = \frac{1}{\sigma}$

④ $\text{Power} = I_{eff}^2 R = \frac{V_{eff}^2}{R} = V_{eff} I_{eff}$

⑤ $V_{eff} = V_{max} \sin(45)$

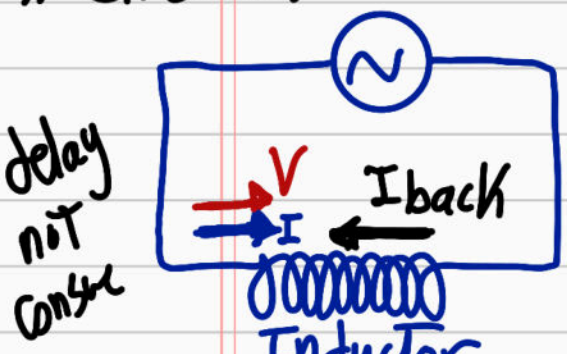
⑥ $I_{eff} = I_{max} \cos(45)$

$\theta = \omega t = 45^\circ$

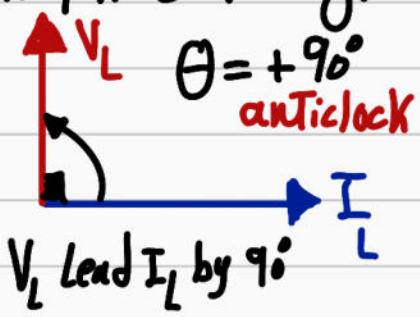


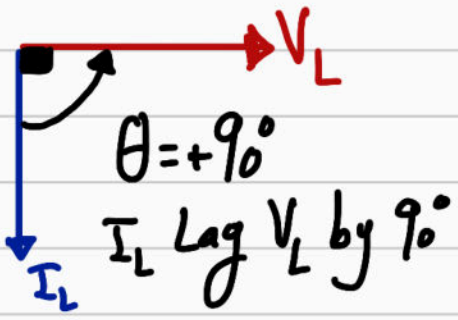
4 Inductive coil of negligable Resistance [L-Circuit] $R=0$

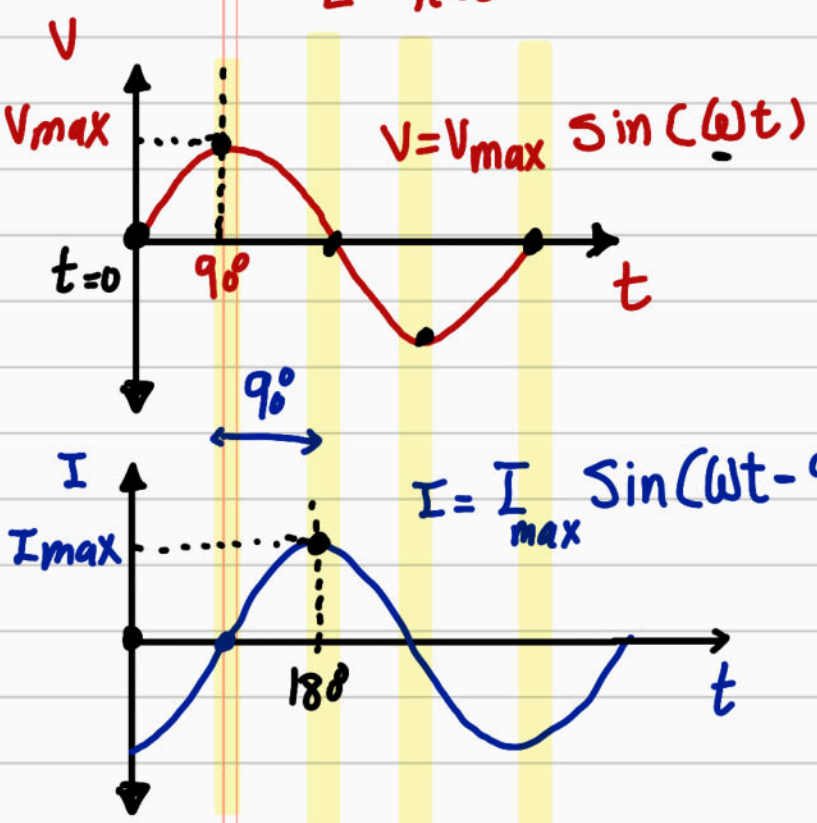
#Circuit:

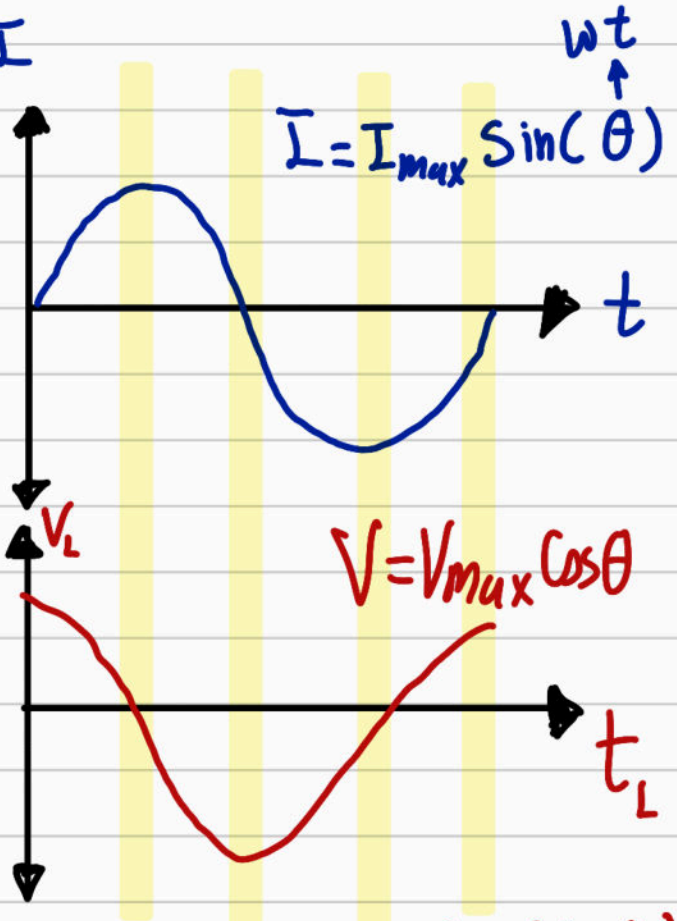


#phasor digram:

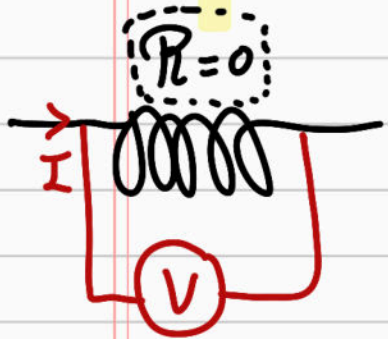








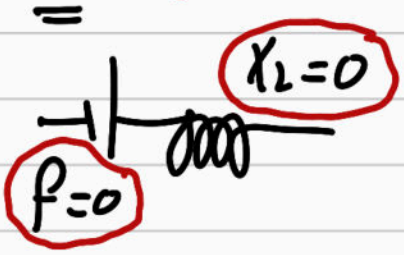
$V=V_{max}\sin(\theta+90)$



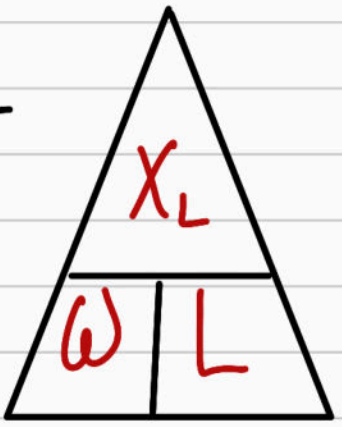
$V_L = I X_L$

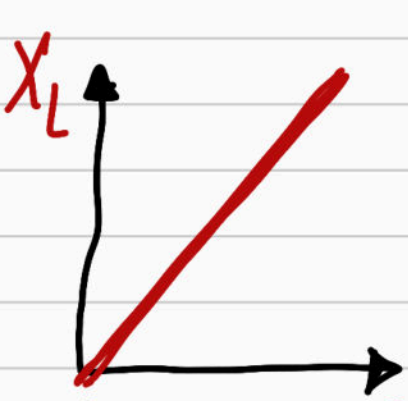
inductive Reactance [Ω]

iT is the opposition To the Flow of Ac due To Self Indution

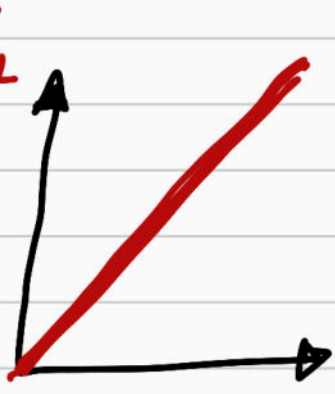


$$X_L = \omega L = 2\pi f L$$



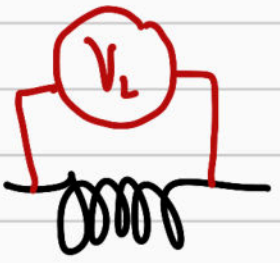


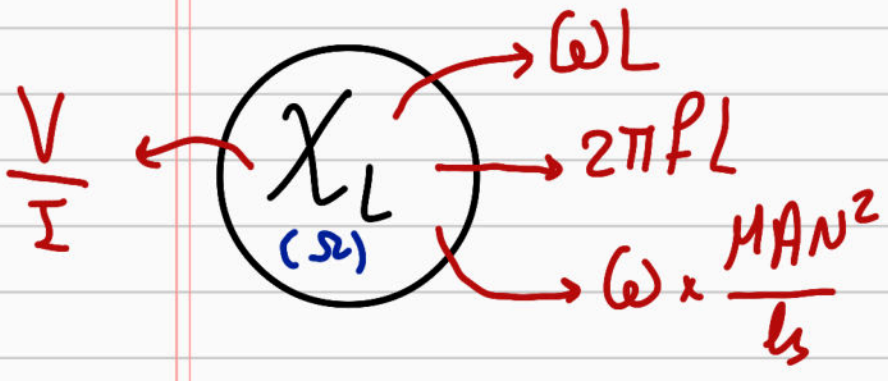
$$\text{Slope} = \frac{X_L}{f} = 2\pi L$$



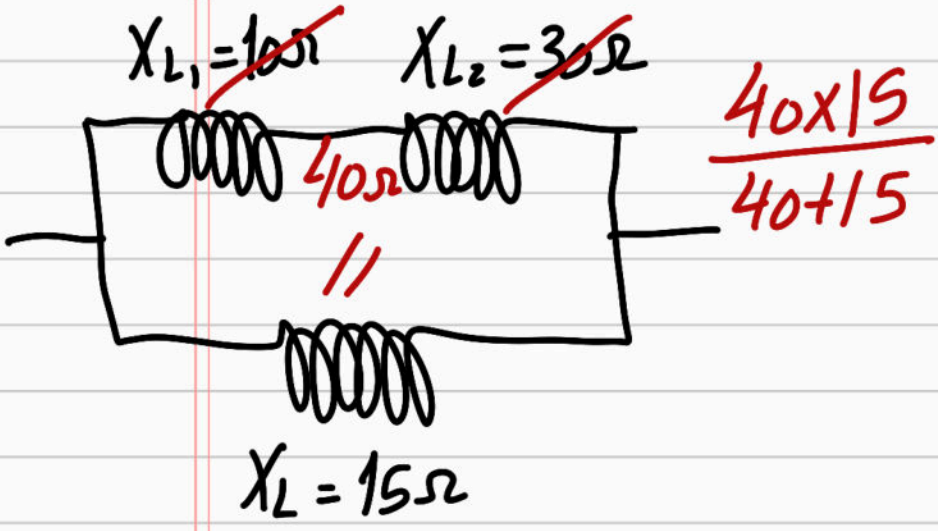
$$\text{Slope} = \frac{X_L}{L} = 2\pi f = \omega$$

Power Consumed = Zero
by Inductor



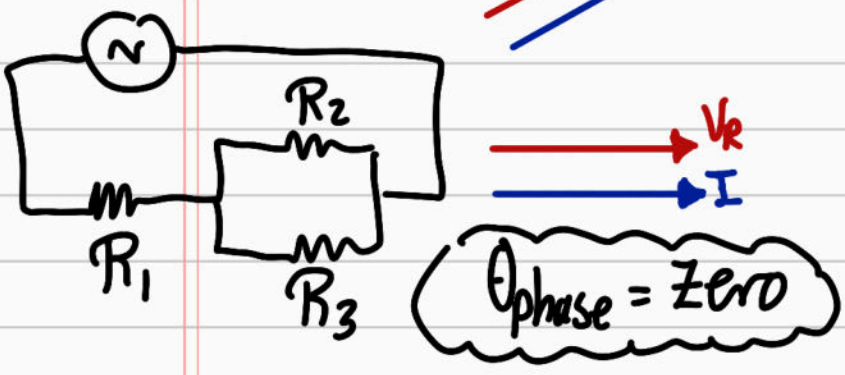


1. $V_L = I X_L$
2. $X_L = \omega L = 2\pi f L$
3. $L = \frac{\mu A N^2}{l_s}$
4. P_W = Zero

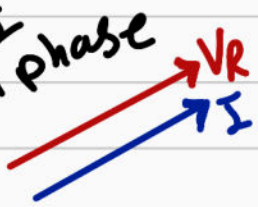


$$\frac{40 \times 15}{40 + 15}$$

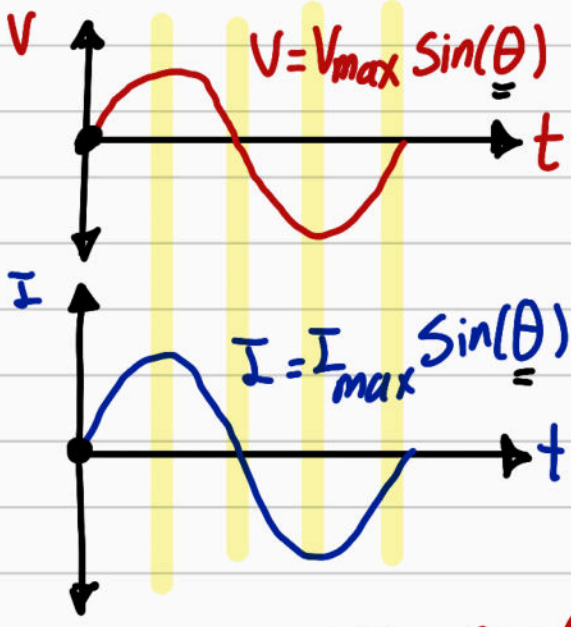
R-Circuit



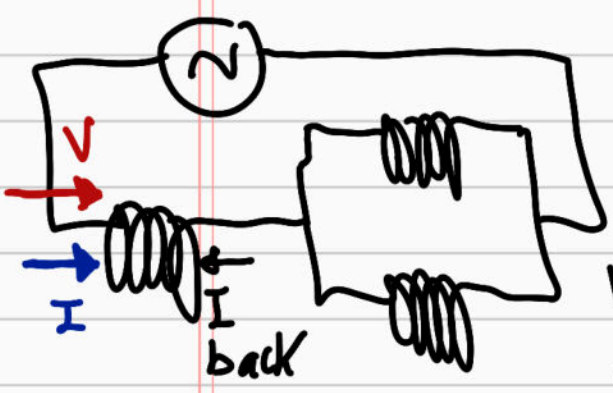
V_R & I in phase

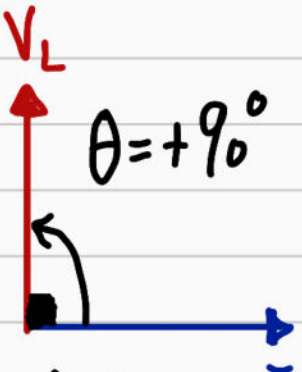


$\theta_{phase} = Zero$



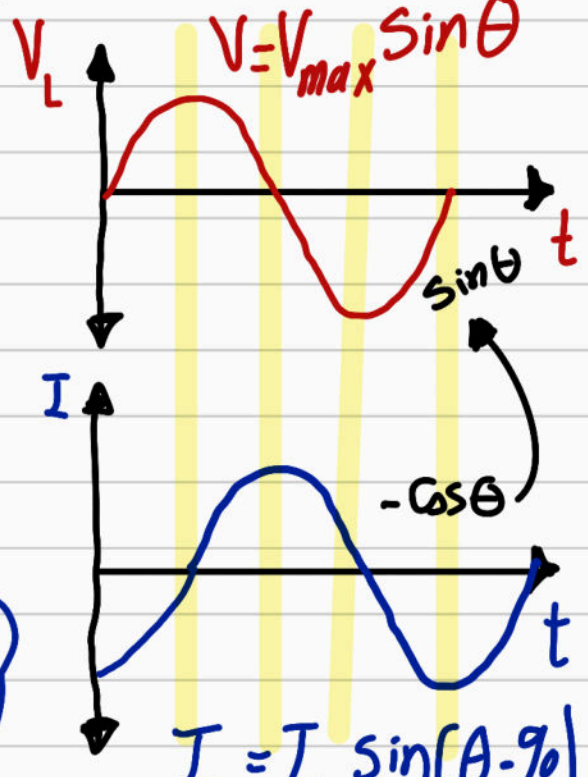
L-Circuit





V_L lead I by 90°

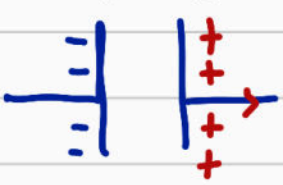
I lag V_L by 90°

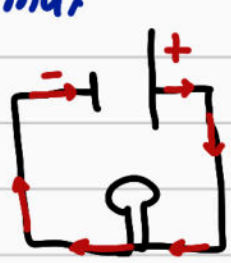


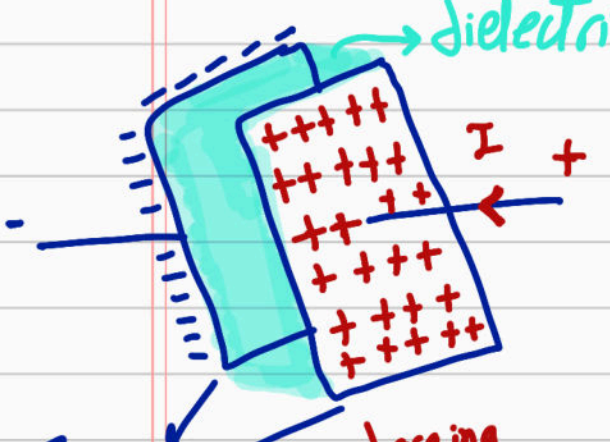
$$\sin(\theta - 90) = -\sin(90 - \theta)$$
$$= -\cos(\theta)$$

$I = I_{max} \sin(\theta - 90)$

#Capacitor [Electric charges store]



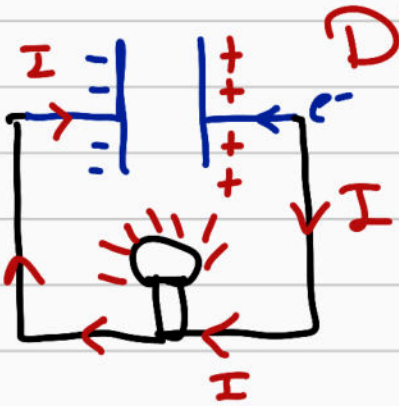


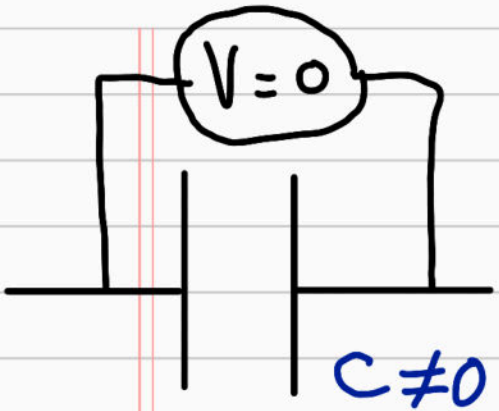


Same Conductor

charging

$Q \propto C$ | $Q \propto V$





C → Capacitance "Capacity" of capacitor

(ظرفية)

Farad [F]

$C \neq 0$

$Q = 0$



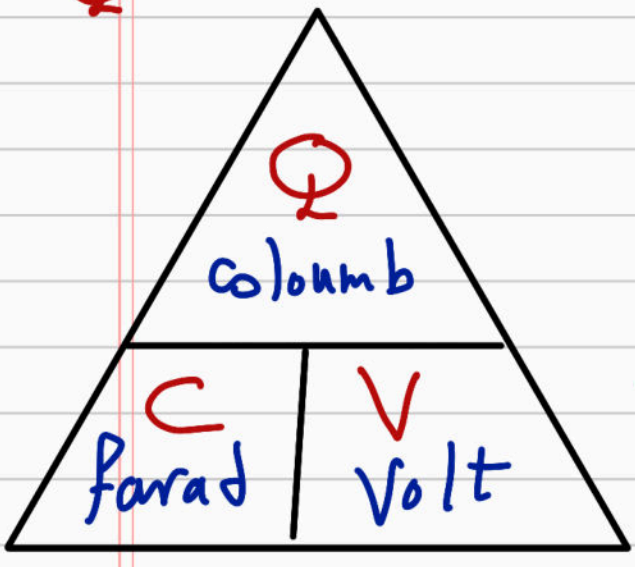
$Q = 0.5 \text{ liter}$

$C = 0.5 \text{ liter}$



$Q = 1 \text{ Ltr}$

$C = 1 \text{ Ltr}$



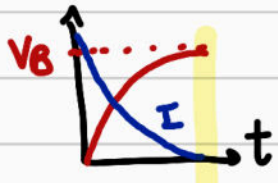
$$\text{farad} = \frac{C}{V} = \frac{As}{V} = \frac{S}{\Omega}$$

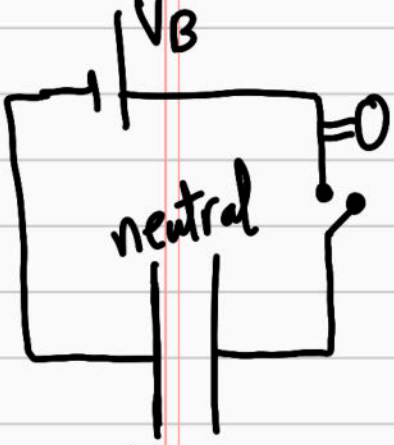
How To charge a Capacitor??

DC Source ✓

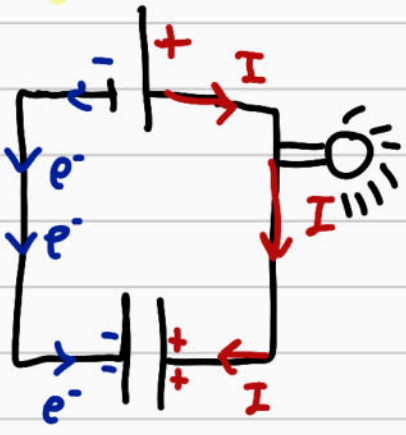
AC Source ✓

Capacitor ✓

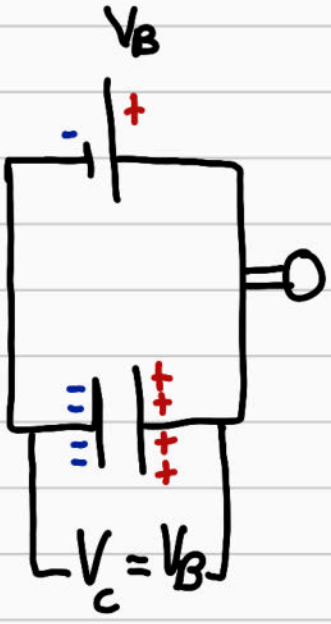




Q=0
V=0
C≠0



Charging the Capacitor

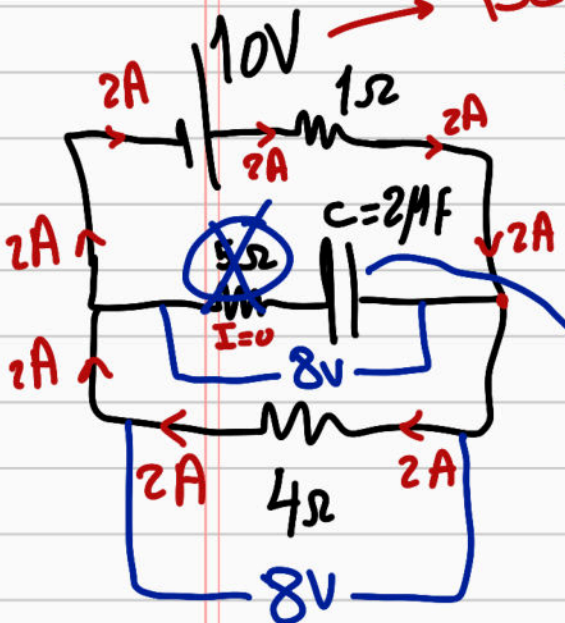


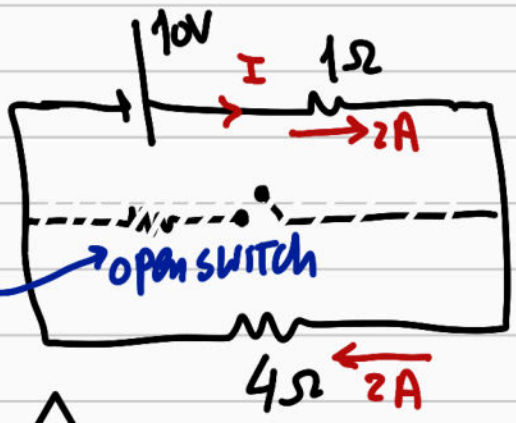
When the Capacitor is fully Charged the lamp Turns off

for Mamy Salma

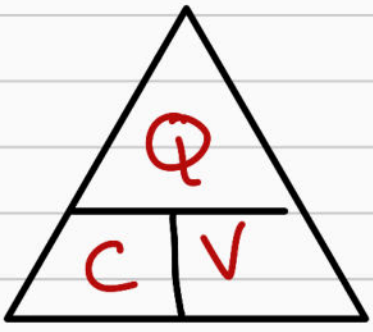
1. DC source [Battery] 2. Capacitor

DC source $\rightarrow f=0 \rightarrow X_C = \frac{1}{2\pi fC} = \frac{1}{0} = \infty$ ↓ open switch





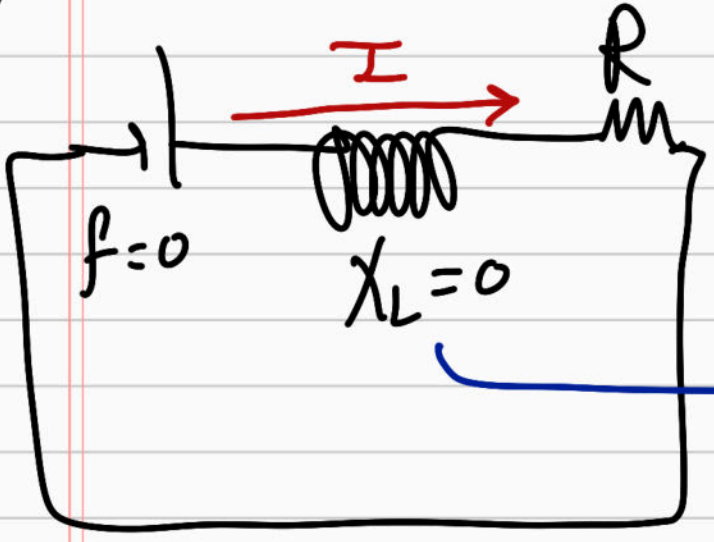
$I = \frac{10}{4+1} = 2A$



$Q = CV = 2 \times 8 = 16\mu C$

for Radwa

1. DC Source ? Indutor



$X_L = 2\pi fL$

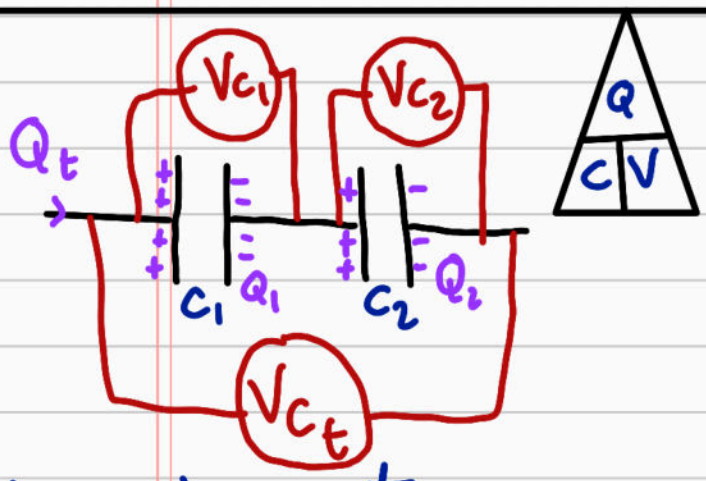
0 0

سلك فاضي



Capacitors Connection

Seires *



V → distribution

$$Vc_t = Vc_1 + Vc_2$$

$$\frac{\cancel{Q_t}}{C_t} = \frac{\cancel{Q_1}}{C_1} + \frac{\cancel{Q_2}}{C_2}$$

Q → constant

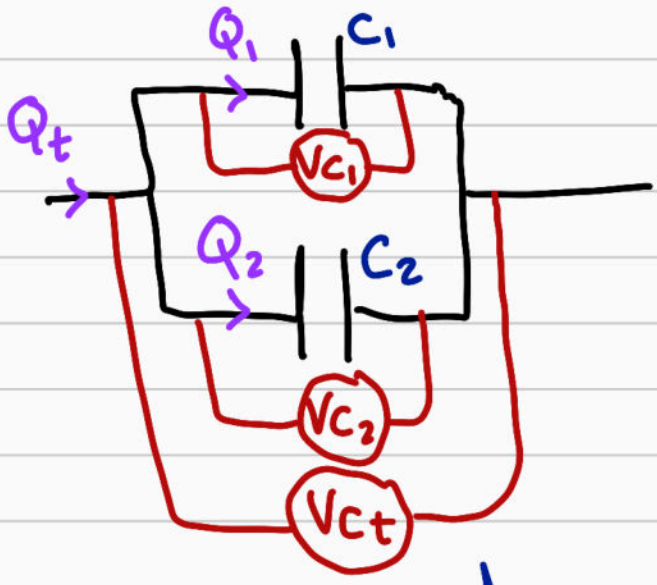
$$Q_t = Q_1 = Q_2$$

$$\frac{1}{C_t} = \frac{1}{C_1} + \frac{1}{C_2}$$ Parallel

$$C_t = \left[\frac{1}{C_1} + \frac{1}{C_2}\right]^{-1}$$

$$C_{eq} = \frac{C_1 \times C_2}{C_1 + C_2}$$

Parallel



Q → distribution

$$Q_t = Q_1 + Q_2$$

$$C_t \cancel{V_t} = C_1 \cancel{V_1} + C_2 \cancel{V_2}$$

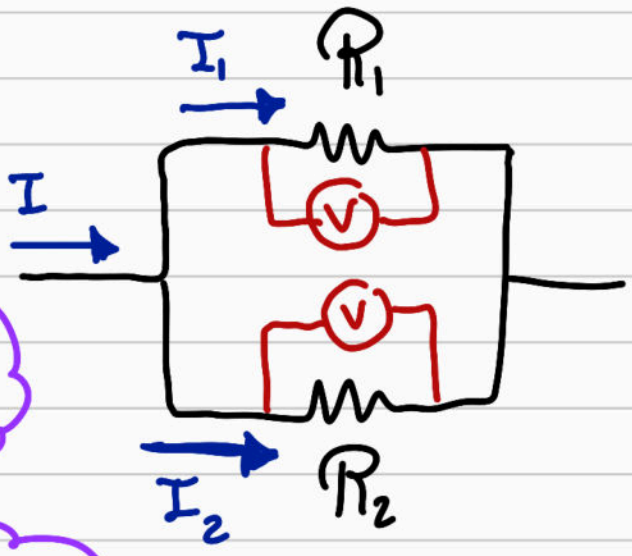
V → constant

$$V_1 = V_2 = V_t$$

$$C_t = C_1 + C_2$$

Seires

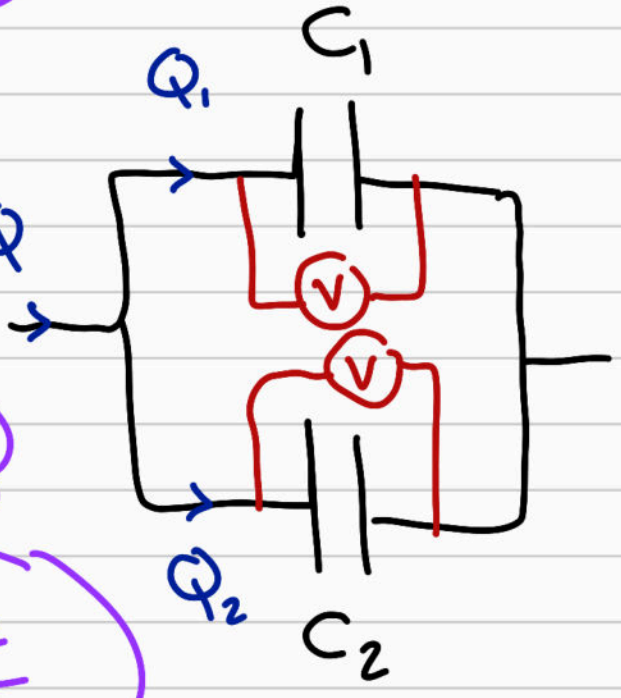
Current divider



$$I = \frac{V}{R}$$

$$I_1 = \frac{I \times R_2}{R_1 + R_2}$$

$$I_2 = \frac{I \times R_1}{R_1 + R_2}$$

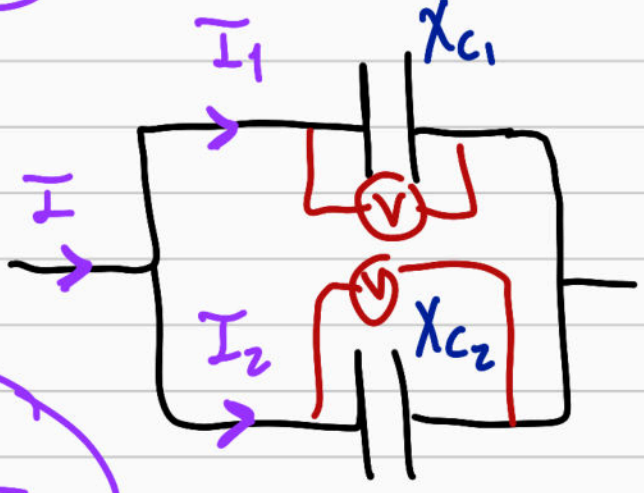


$$Q = CV$$

constant

$$Q_1 = \frac{Q \times C_1}{C_1 + C_2}$$

$$Q_2 = \frac{Q \times C_2}{C_1 + C_2}$$

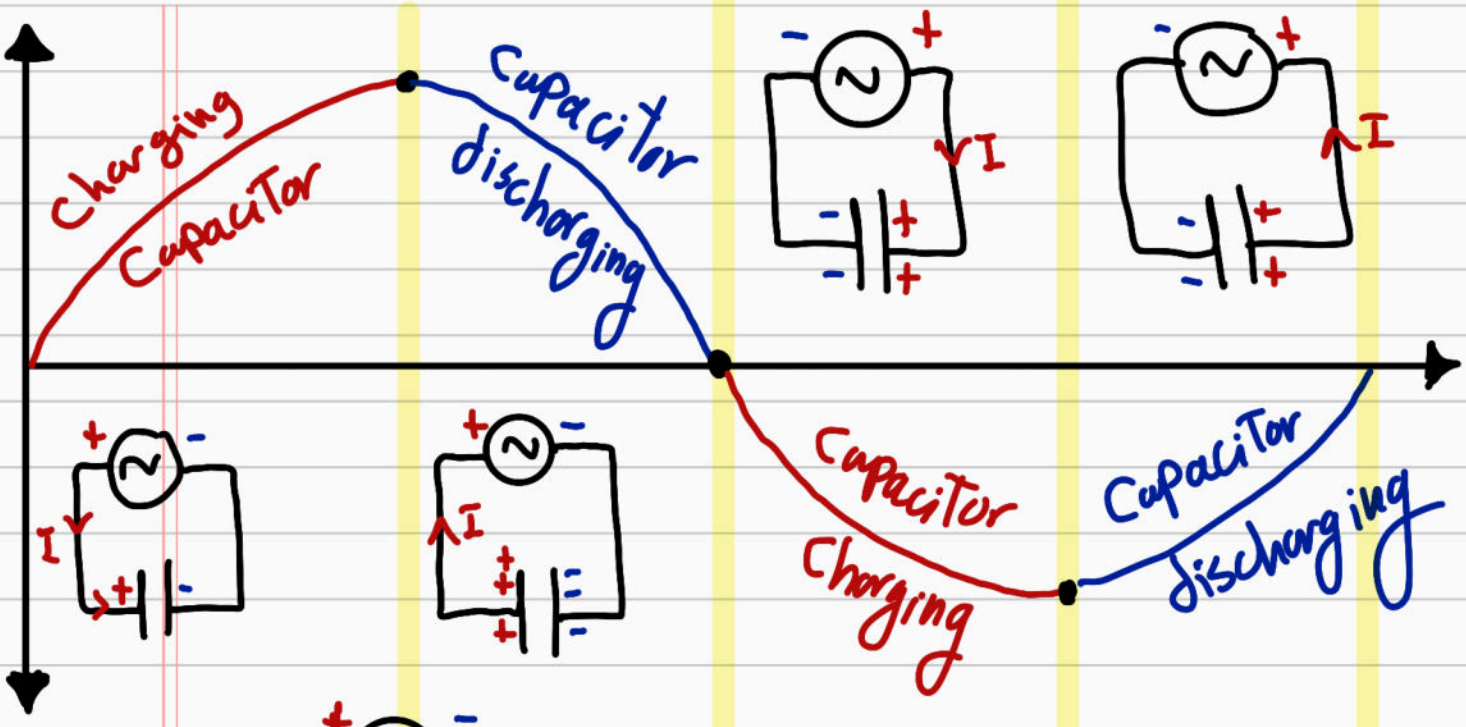


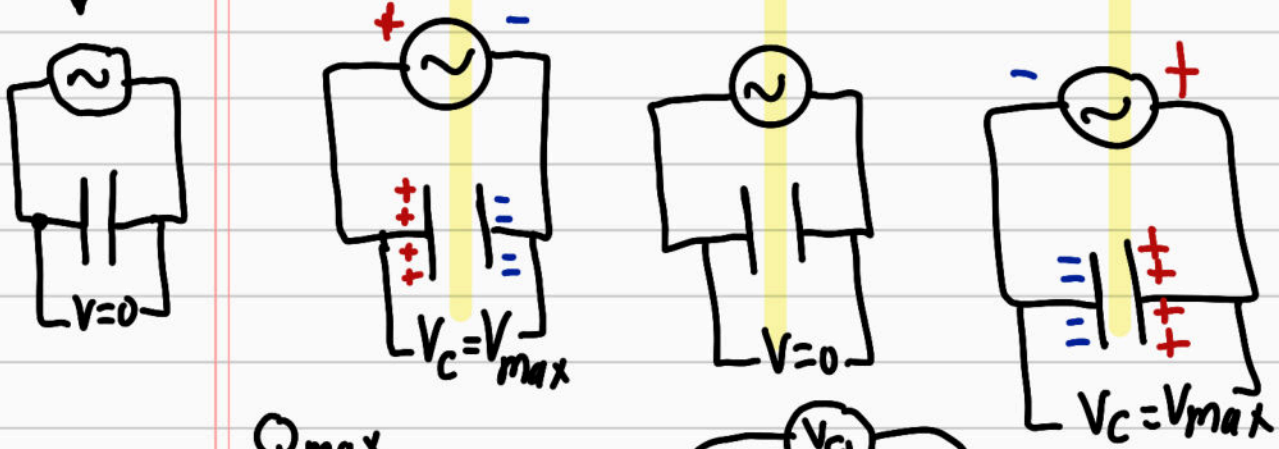
$$I = \frac{V}{X_C}$$

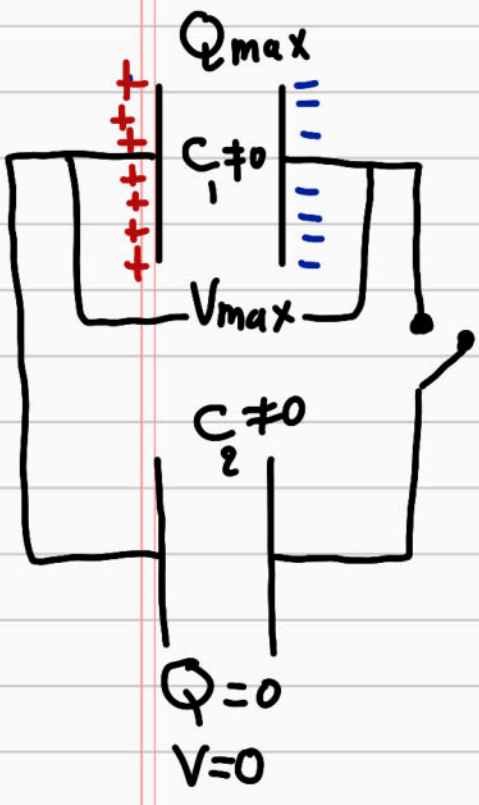
const

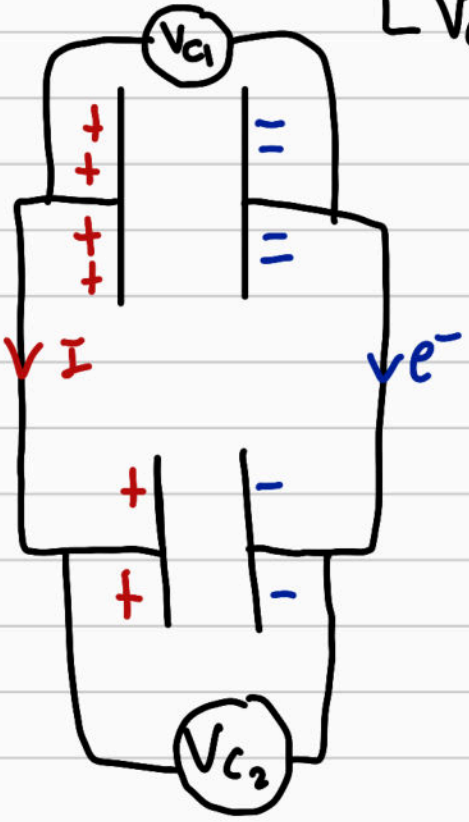
$$I_1 = \frac{I \times X_{C_2}}{X_{C_1} + X_{C_2}}$$

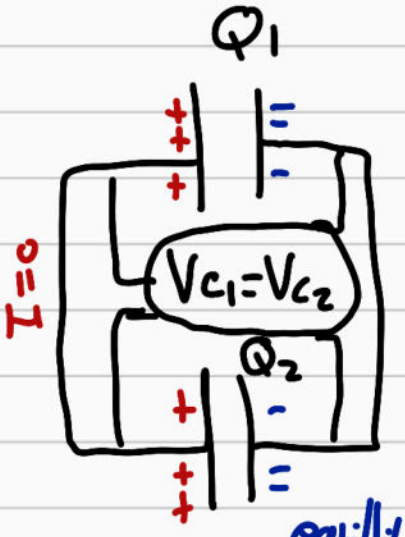
$$I_2 = \frac{I \times X_{C_1}}{X_{C_1} + X_{C_2}}$$







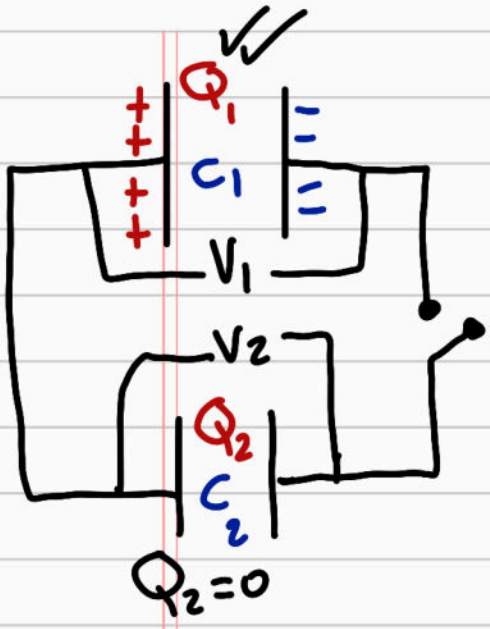


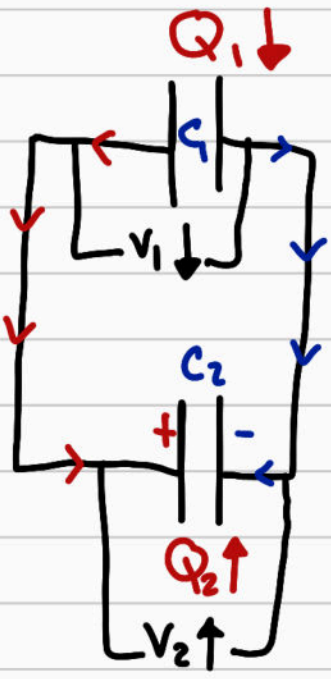


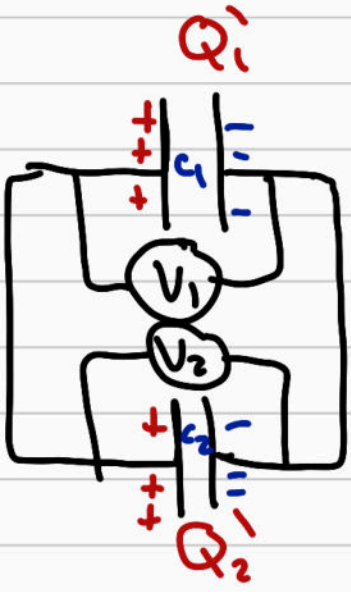
$$V_{C_1} = V_{C_2}$$

$$\frac{Q_1}{C_1} = \frac{Q_2}{C_2}$$

$$Q_{max} = Q_1 + Q_2$$







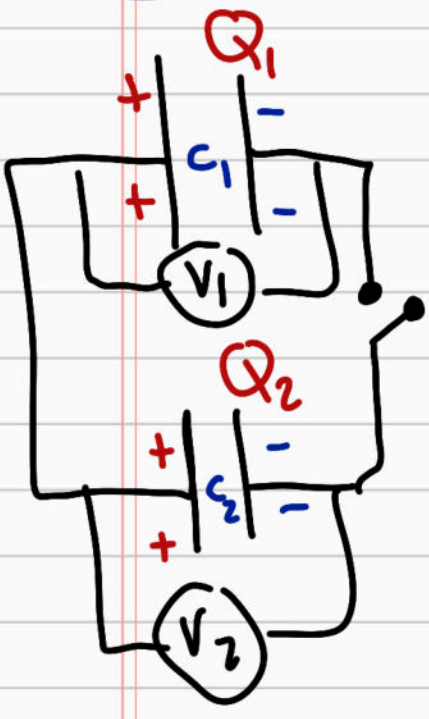
$V_1 = V_2$

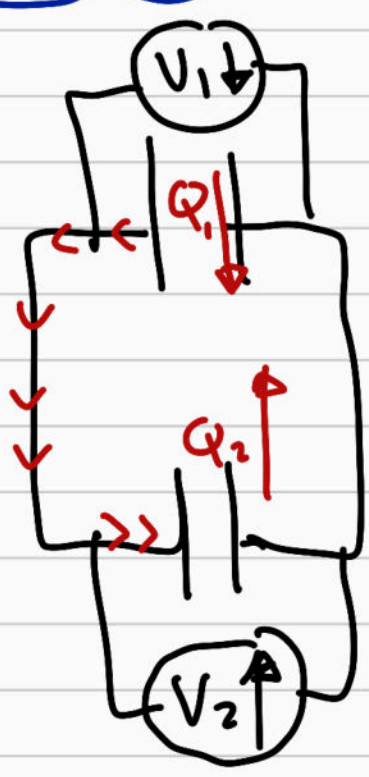
Before After

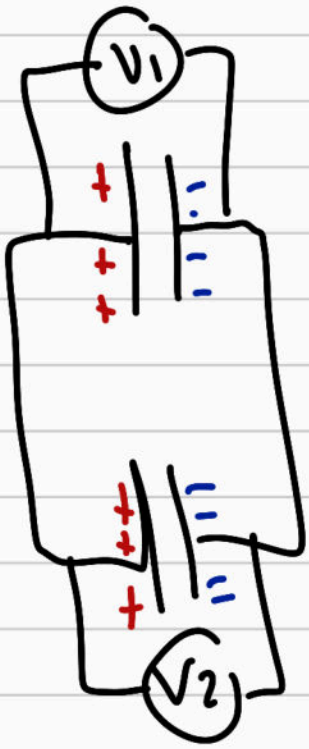
$Q_1 + Q_2 = Q_1' + Q_2'$

$Q_2 \to 0$

$\frac{Q_1'}{C_1} = \frac{Q_2'}{C_2}$





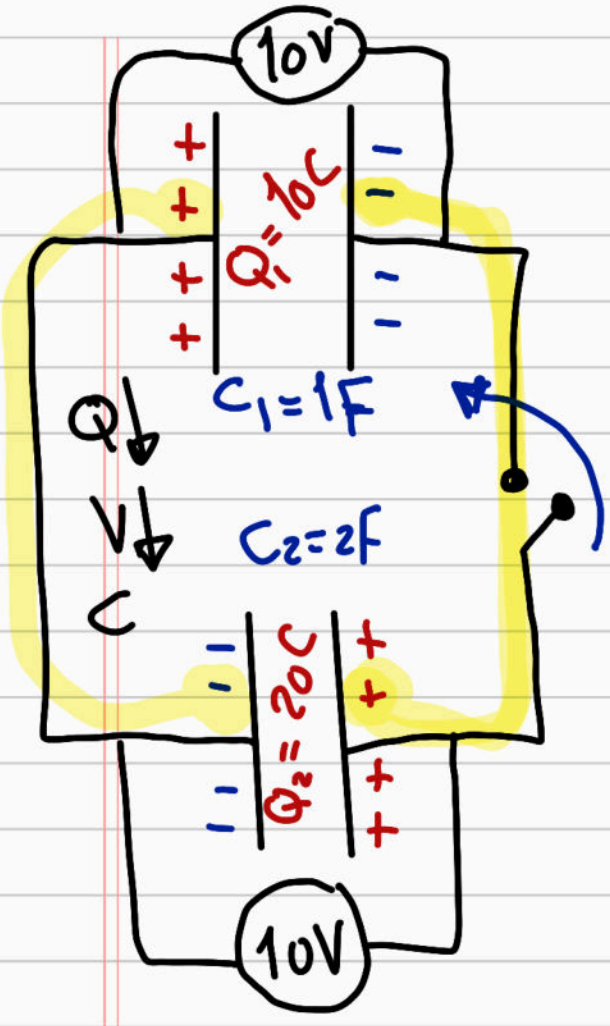


$Q_1 + Q_2 = Q_1' + Q_2'$

Before After

$\frac{Q_1'}{C_1} = \frac{Q_2'}{C_2}$

$V_1 = V_2$



Before

$$Q_1 + Q_2 = Q_1' + Q_2'$$

$$+20 + (-10) = Q_1' + Q_2'$$

$$Q_1' + Q_2' = 10$$

$$\frac{Q_1'}{1} = \frac{Q_2'}{2}$$

$$Q_2' = 2Q_1'$$

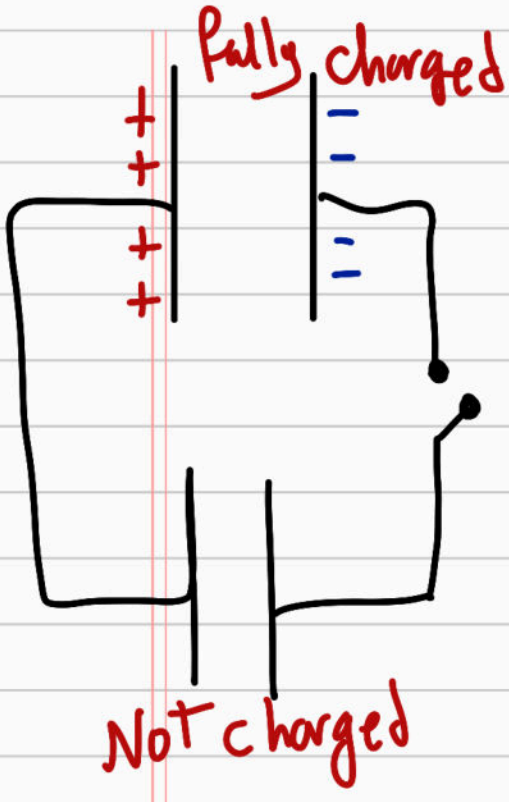
$$Q_1' + 2Q_1' = 10$$

$$3Q_1' = 10 \rightarrow Q_1' = \frac{10}{3}C$$

$$Q_2' = 2Q_1' \rightarrow Q_2' = \frac{20}{3}C$$

$$V_1 = \frac{Q_1'}{C_1} = \frac{10/3}{1} = \frac{10}{3} = 3.333V \quad \text{dee.}$$

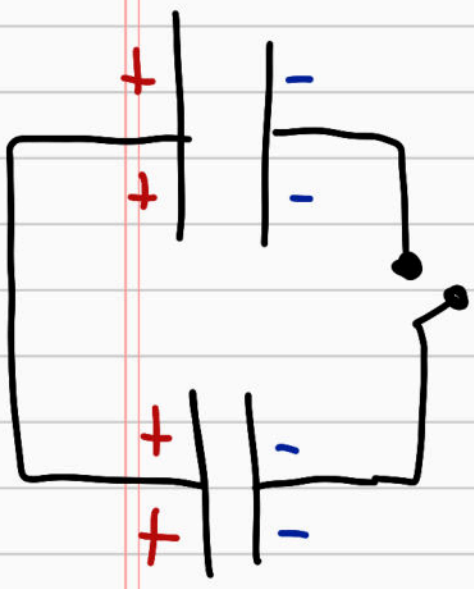
$$V_2 = \frac{Q_2'}{C_2} = \frac{20/3}{2} = \frac{10}{3} = 3.333V \quad \text{dee.}$$



$$Q_1 = Q_1' + Q_2'$$

$$V_1 = V_2$$

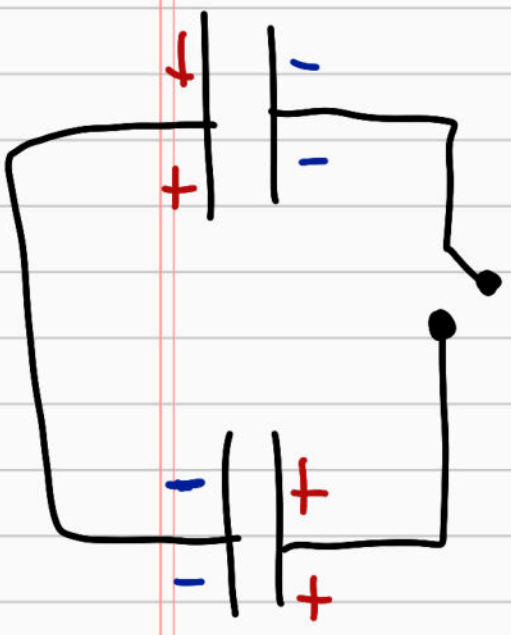
$$\frac{Q_1'}{C_1} = \frac{Q_2'}{C_2}$$



$$Q_1 + Q_2 = Q_1' + Q_2'$$

$$V_1 = V_2$$

$$\frac{Q_1'}{C_1} = \frac{Q_2'}{C_2}$$



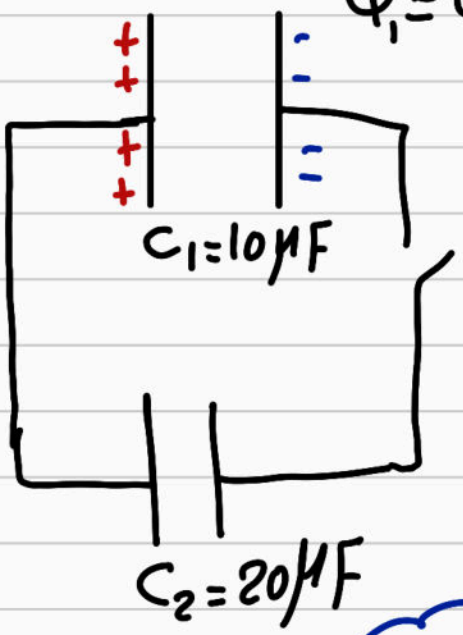
$$Q_1 - Q_2 = Q_1' + Q_2'$$

$$V_1 = V_2$$

$$\frac{Q_1'}{C_1} = \frac{Q_2'}{C_2}$$

Example

2023
1st session



Find Q_1 & Q_2 after closing the switch??

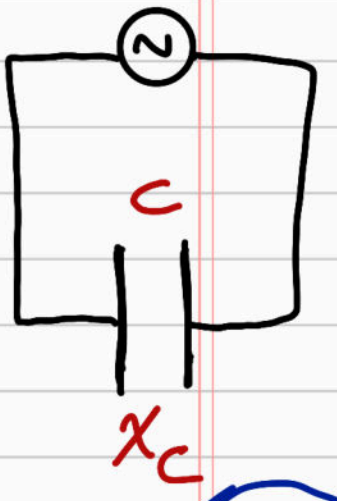
$$Q_1 + Q_2 = 60$$

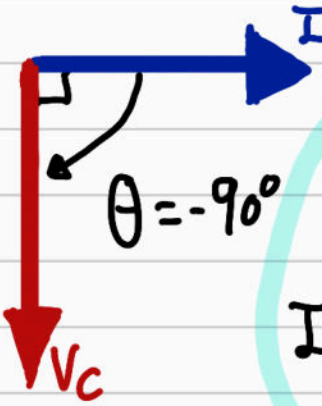
$$\frac{Q_1}{C_1} = \frac{Q_2}{C_2}$$

$$Q_1 = \frac{1}{2} Q_2$$

$Q_1 = 20\mu C$ $Q_2 = 40\mu C$

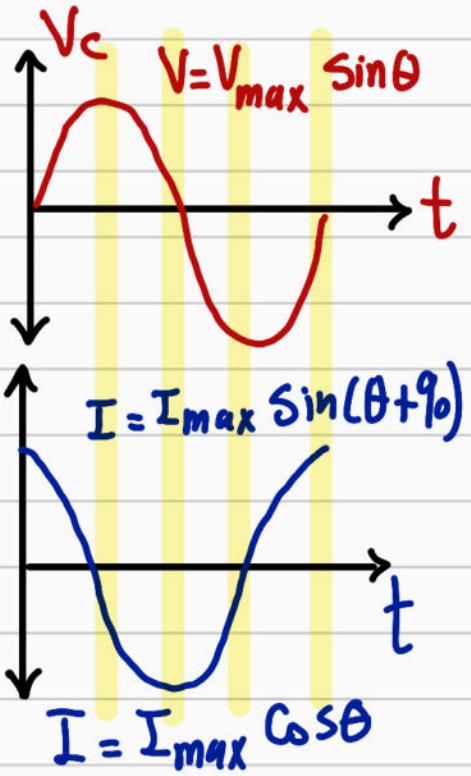
*C-Circuit: Capacitor circuit





$\theta = -90°$ V_C lags I by 90°

I leads V_C by 90°

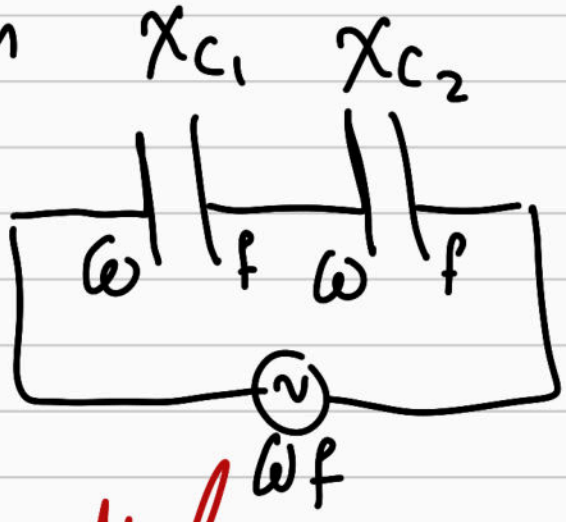


$$\sin(\theta + 90) = \sin(90 + \theta) = \cos\theta$$

#Rules

1. $V_C = I X_C$ → Capacitive Reactance [Ω]
2. $X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$
3. $C = \frac{Q}{V}$
4. P_w = zero

Capacitive Reactance Connection



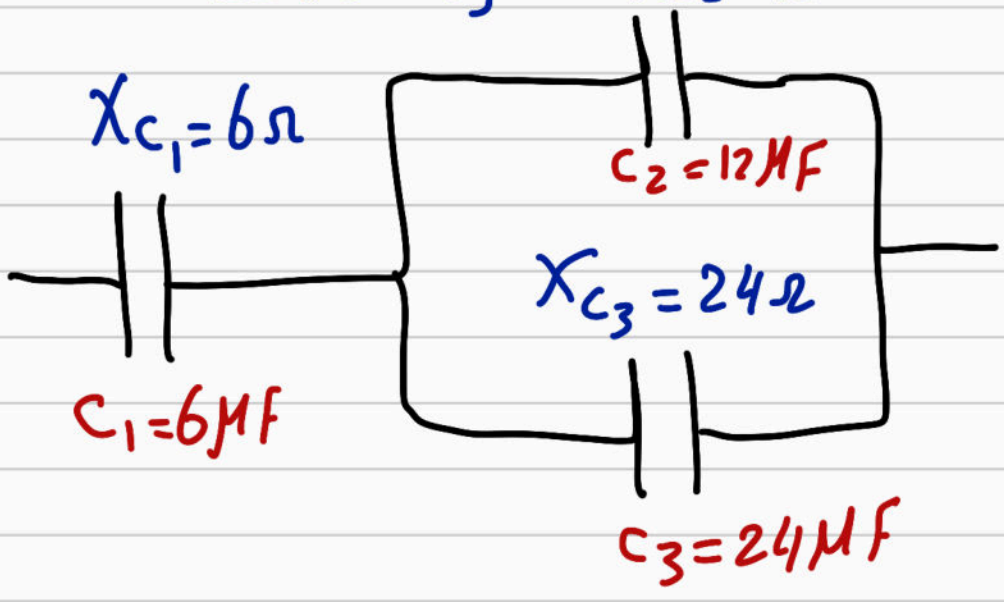
$$X_{C_{eq}} = X_{C_1}^{(\Omega)} + X_{C_2}^{(\Omega)}$$

$$\frac{1}{\omega C_{eq}} = \frac{1}{\omega C_1} + \frac{1}{\omega C_2}$$

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} \rightarrow \text{Parallel}$$

$C_2 \rightarrow$ [Farad]

Capacitance of capacitor



$$X_{C_{eq}} = \frac{24 \times 12}{24+12} + 6 = 14\,\Omega$$

$$C_{eq} = \frac{6 \times 36}{6+36} = \frac{36}{7}\,\mu F$$

$$\tilde{I} = \frac{V_{source}}{X_{C_{eq}}} \rightarrow X_{C_{eq}} = \frac{1}{2\pi f C_{eq}}$$

$$\sqrt{\frac{L}{C}} = \sqrt{\frac{\Omega \cdot S}{\frac{C}{V}}}$$

L: self inductance

C: Capacitance

$$H = \frac{V \cdot S}{A}$$

$$H = \Omega \cdot S$$

$$L\Delta I = N\Delta\phi$$

$$H \cdot A = Wb$$

$$H = \frac{Wb}{A}$$

$$emf = N\frac{\Delta\phi}{\Delta t}$$

$$V = \frac{Wb}{S}$$

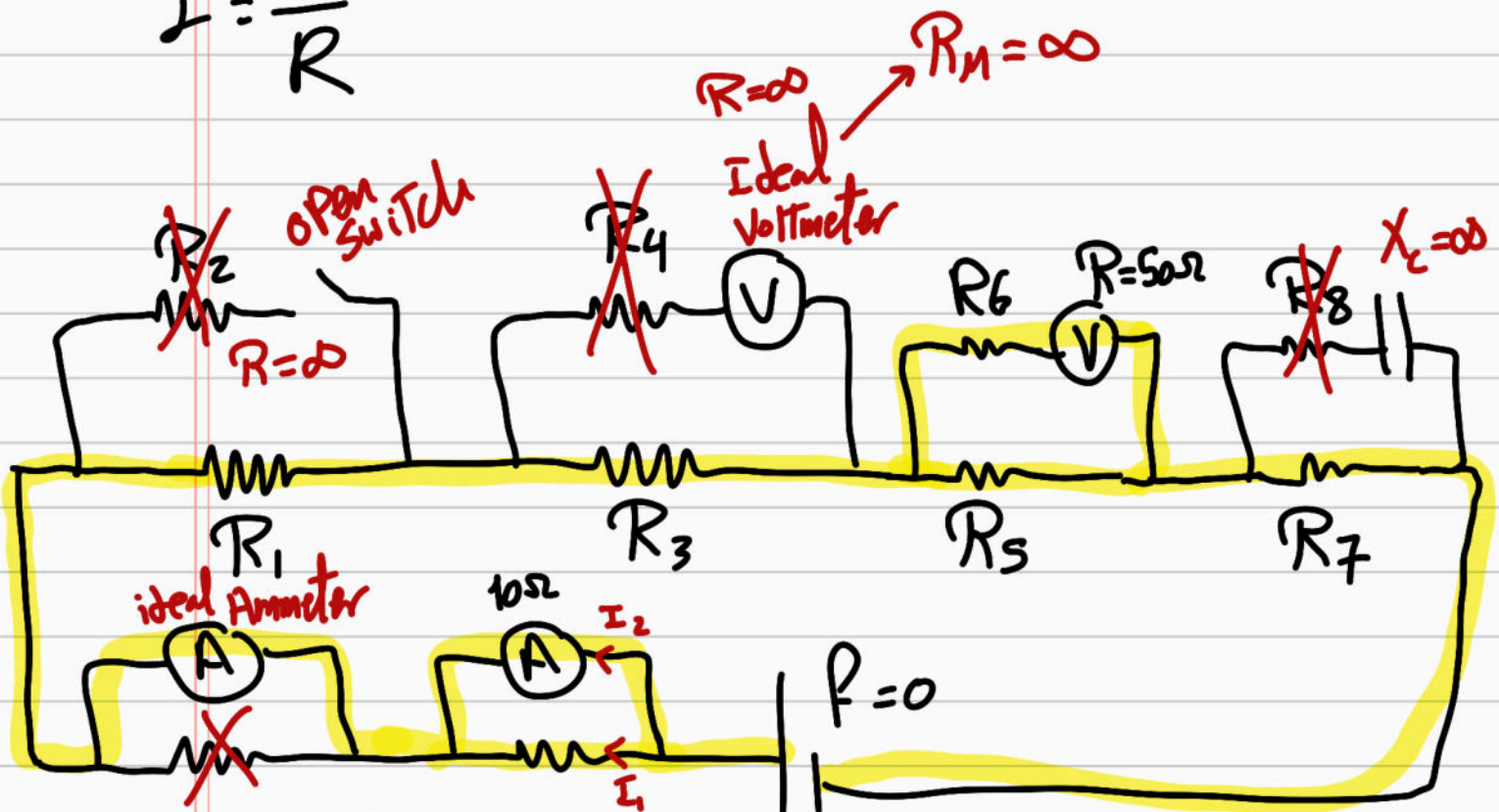
$$\boxed{Wb = V \cdot S}$$

$$C = \frac{Q}{V} = \sqrt{\frac{\Omega \cdot S \cdot V}{C}} = \sqrt{\frac{\Omega V}{A}} = \Omega \cdot \Omega$$

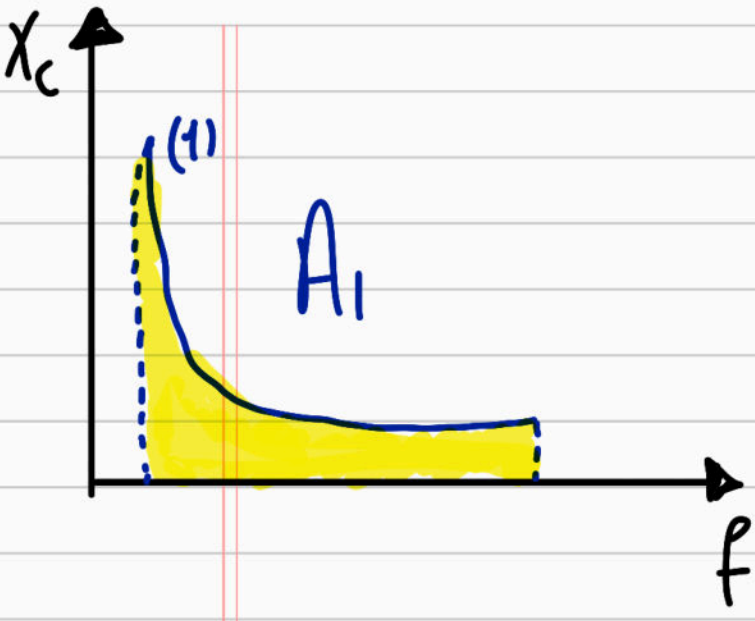
$$= \sqrt{\Omega^2} = \Omega \rightarrow R$$

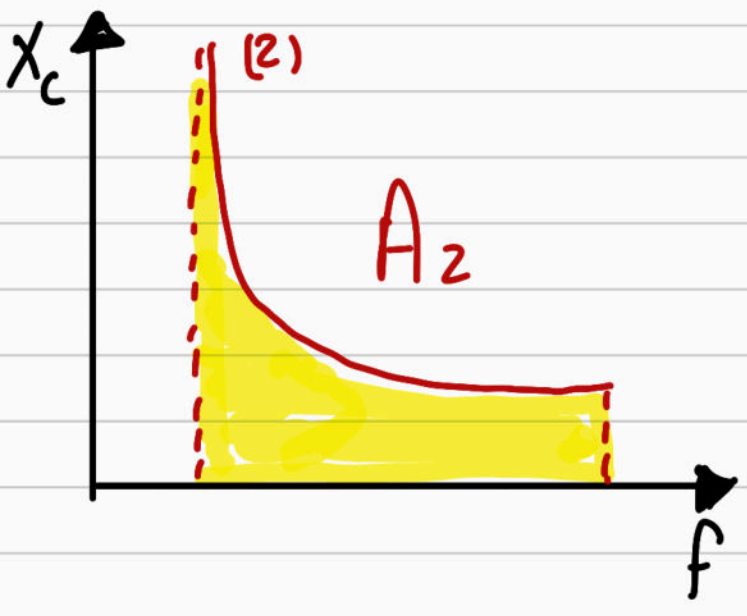
$$CR \rightarrow \frac{Q}{V} \times R = \frac{QR}{V} = \frac{Q}{I} = t \rightarrow \text{second.}$$

$$I = \frac{V}{R}$$



$$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi \times 0 \times C} = \frac{1}{0} = \infty$$





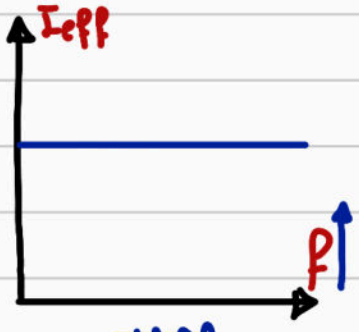
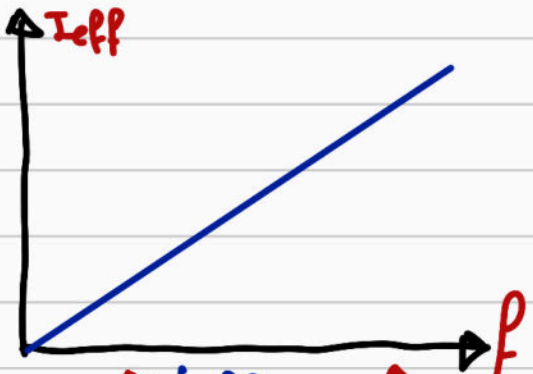
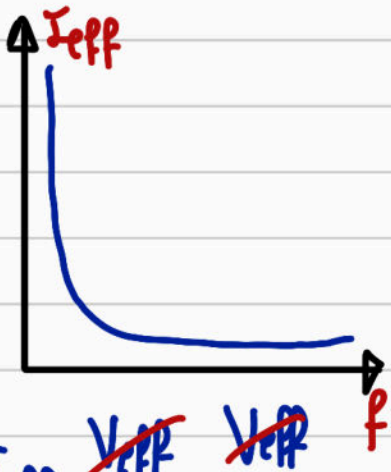
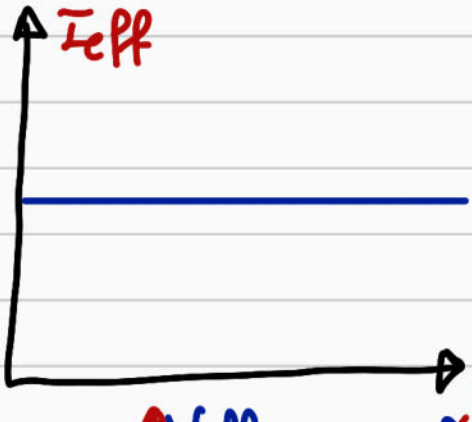
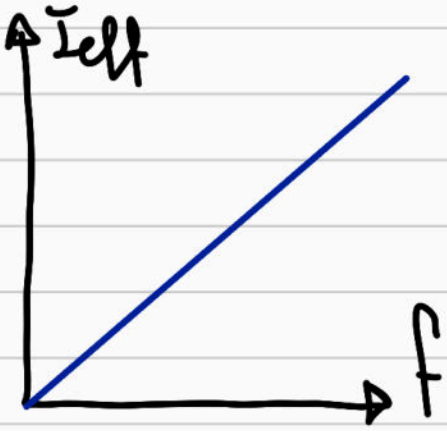
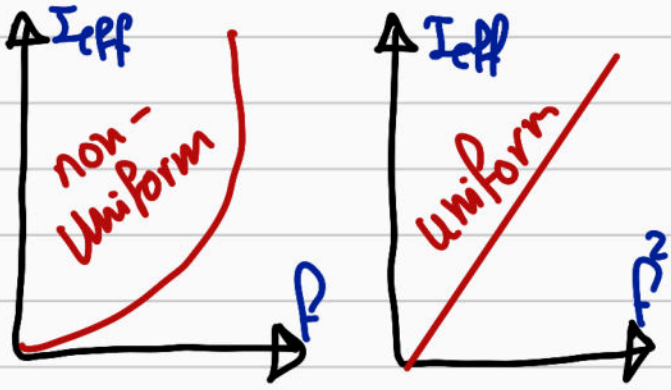
Area under curve = y-axis x x-axis

$$= X_C \times f = \frac{1}{2\pi C} = \text{Area under curve}$$

$A_1 < A_2$

$C_1 > C_2$

موفق باشی
2026

Circuit	AC Source doesn't ~~depend~~ on frequeny	AC Dynamo depend on frequeny
R	 $I_{eff} = \frac{V_{eff}}{R} \rightarrow R = \frac{\rho L}{A}$	 $\uparrow I_{eff} = \frac{\uparrow V_{eff}}{R} = \frac{BA2\pi\cancel{f}N}{R\sqrt{2}}$
L	 $\downarrow I_{eff} = \frac{\cancel{V_{eff}}}{X_L\uparrow} = \frac{\cancel{V_{eff}}}{2\pi\cancel{f}L}$	 $I_{eff} = \frac{\uparrow V_{eff}}{X_L\uparrow} = \frac{BA2\pi\cancel{f}N}{2\pi\cancel{f}L\sqrt{2}}$
C	 $\uparrow I_{eff} = \frac{V_{eff}}{X_C\downarrow} = \frac{V_{eff}}{\frac{1}{2\pi f C}}$ $= V_{eff} \times 2\pi f\uparrow C$	 $I_{eff} = \frac{V_{eff}}{X_C} = \frac{BA2\pi fN}{\frac{1}{2\pi fC}\sqrt{2}}$ $I_{eff} \propto f^2$

R - Circuit

V_R & I_R in phase

$I = I_{max} \sin(\omega t)$

$V = V_{max} \sin(\omega t)$

$V = V_{max} \sin(\omega t + 90^\circ)$

$I = I_{max} \sin(\omega t + 90^\circ)$

L - Circuit

$\pi/2$

V_L lead I by 90°

$I = I_{max} \sin(\omega t)$

$V = V_{max} \sin(\omega t + 90^\circ)$

I lag V_L by 90°

$V = V_{max} \sin(\omega t)$

$I = I_{max} \sin(\omega t - 90^\circ)$

C - Circuit

V_C lag I by 90°

$I = I_{max} \sin(\omega t)$

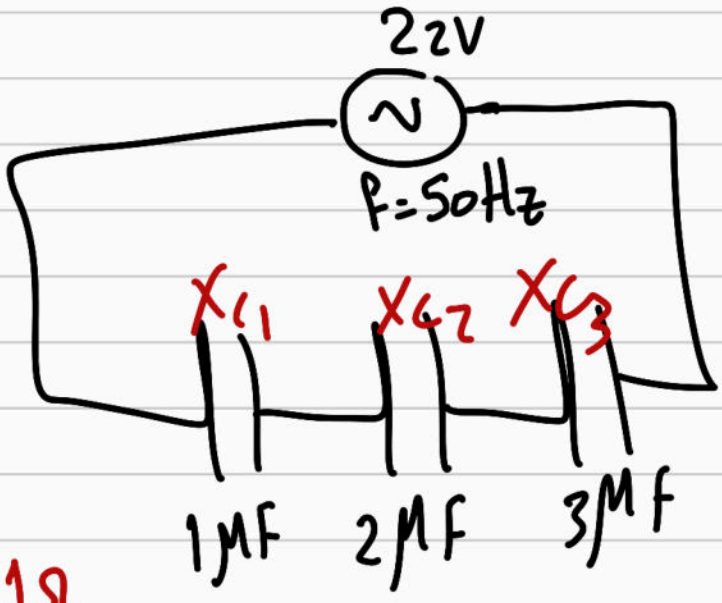
$V = V_{max} \sin(\omega t - 90^\circ)$

I lead V_C by 90°

$V = V_{max} \sin(\omega t)$

$I = I_{max} \sin(\omega t + 90^\circ)$

$$X_{C_1} = \frac{1}{2\pi f C_1}$$



$$X_{C_1} = \frac{1}{2\pi \times 50 \times 1 \times 10^{-6}} = 3183.1\Omega$$

$$X_{C_2} = \frac{1}{2\pi \times 50 \times 2 \times 10^{-6}} = 1591.5\Omega$$

$$X_{C_3} = \frac{1}{2\pi \times 50 \times 3 \times 10^{-6}} = 1061.03\Omega$$

$$X_{C_{eq}} = X_{C_1} + X_{C_2} + X_{C_3} = 5835.63\Omega$$

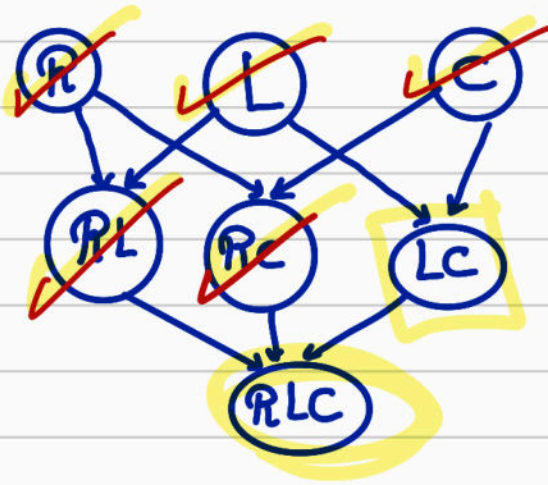
$$I_{max} = \frac{V}{X_{C_{eq}}} = \frac{2}{5835.63} = 3.76 \times 10^{-3} A$$

$$V_{C_1} = I X_{C_1} = 3.76 \times 10^{-3} \times 3183.1 = 12V$$

$$V_{C_2} = I X_{C_2} = 3.76 \times 10^{-3} \times 1591.5 = 6V$$

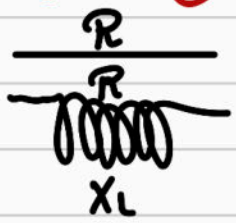
$$V_{C_3} = I X_{C_3} = 3.76 \times 10^{-3} \times 1061.03 = 4V$$

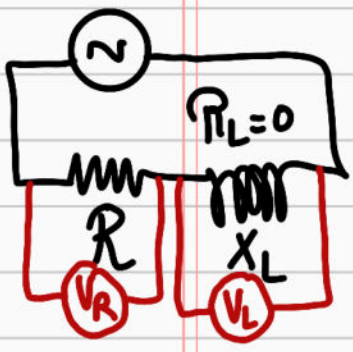
Lesson ②

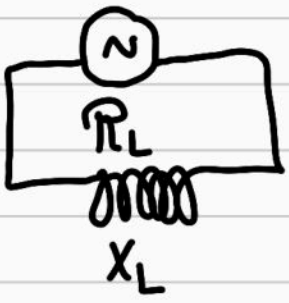


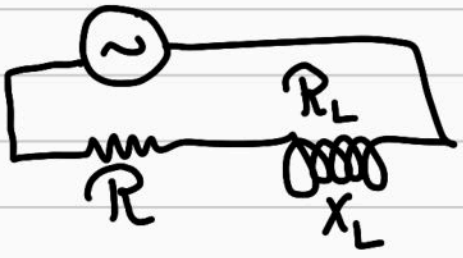
1 RL - Circuit

- ohmic Resistance
- Inductive coil





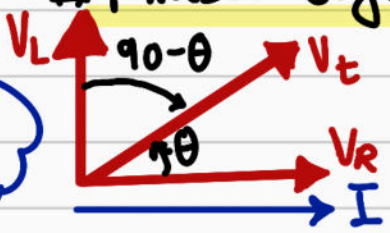






$0° < \theta < 90°$

#Phasor digram



V_t Lead V_R by θ
V_t Lead I by θ
V_t Lags V_L by 90-θ

$V_R = V_{max} \sin(\omega t)$



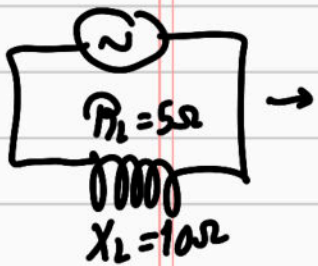
$I = I_{max} \sin(\omega t)$

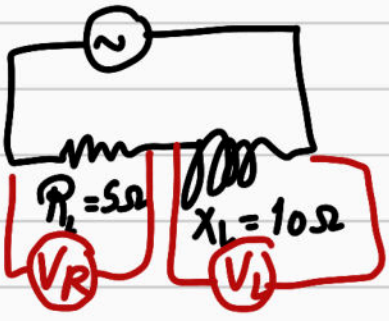
$V_t = V_{max} \sin(\omega t + \theta)$

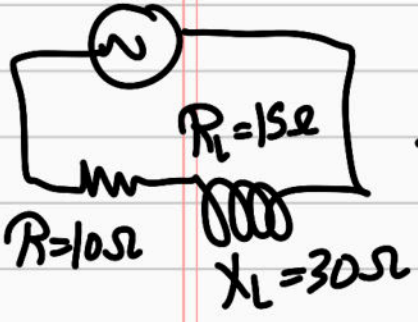
$V_L = V_{max} \sin(\omega t + 90°)$

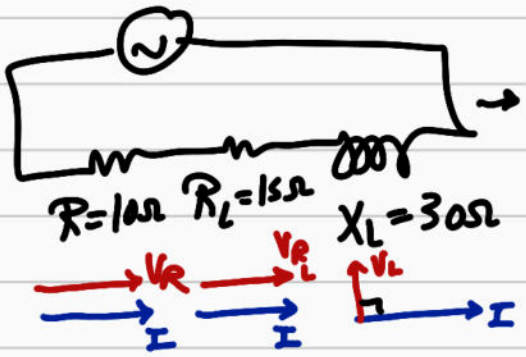
$emf = emf \sin(\theta)$ ← 360 of
instant. max → $\theta = \omega t$

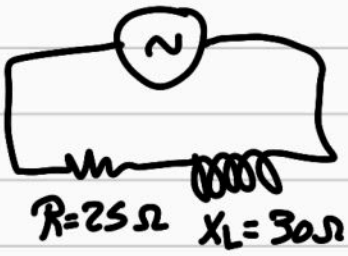
Example s

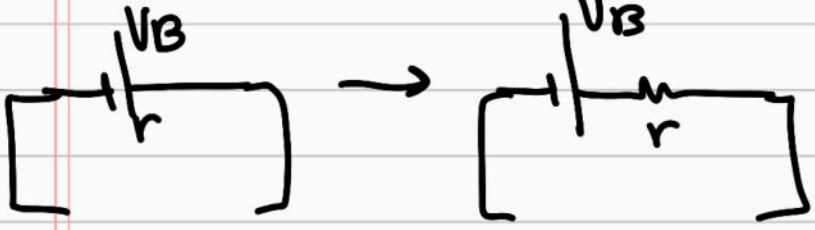












Rules:

① $V_t = \sqrt{V_R^2 + V_L^2}$

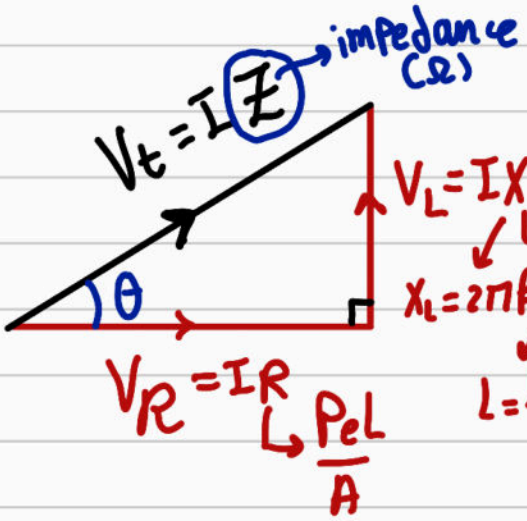
② $\tan\theta = \frac{V_L}{V_R}$

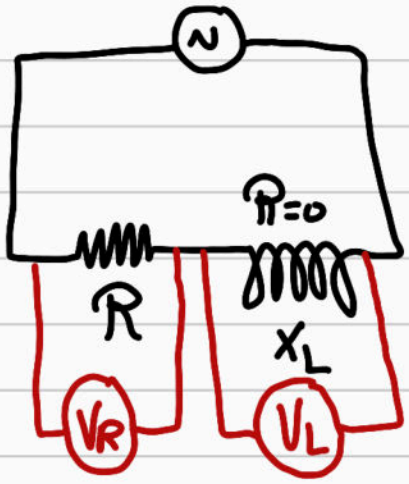
$\sin\theta = \frac{V_L}{V_t}$ $\cos\theta = \frac{V_R}{V_t}$

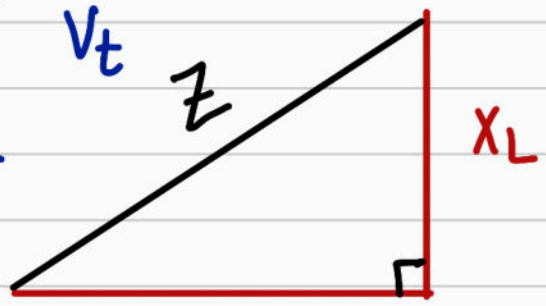
③ $Z = \sqrt{R^2 + X_L^2}$

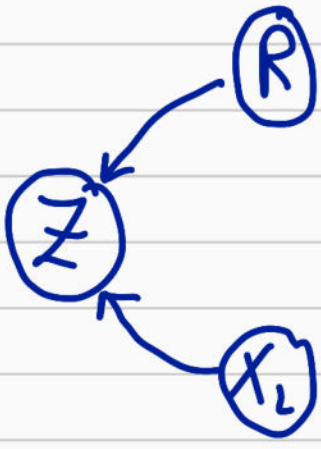
④ $V_t = I Z$

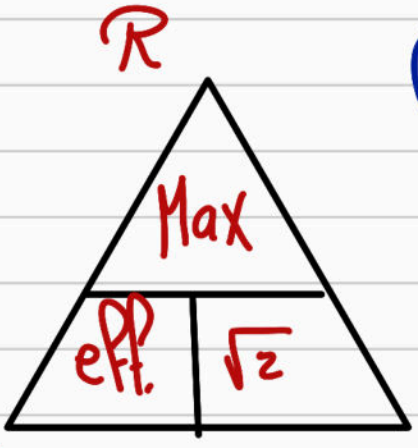
⑤ $P_W = I_{eff}^2 R$











$P_W = \frac{(V_R)_{eff}^2}{R}$

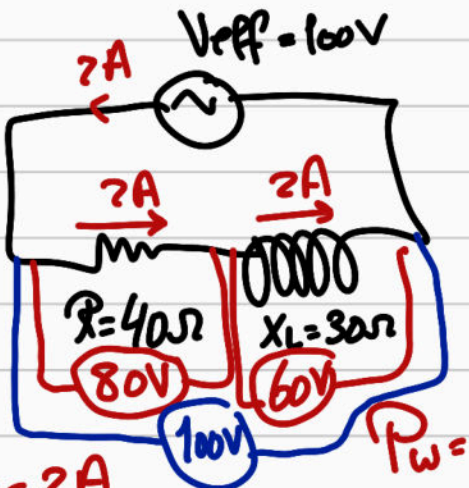
1. effective
2. V_R, R

NOT V_t

$P_W = 100 \times 2 = 200\,W$

$P_W = \frac{100^2}{10}$

Example ①



$Z = \sqrt{R^2 + X_L^2}$

$= \sqrt{40^2 + 30^2} = 50\Omega$

$I_{eff} = \frac{V_t}{Z} = \frac{100}{50} = 2A$

$P_W = I^2 R$

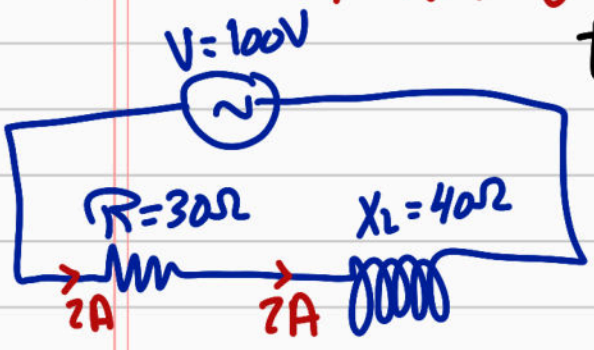
$= (2)^2 \times 40$

$= 160\,W$

$P_W = \frac{80^2}{40} = 160\,W$

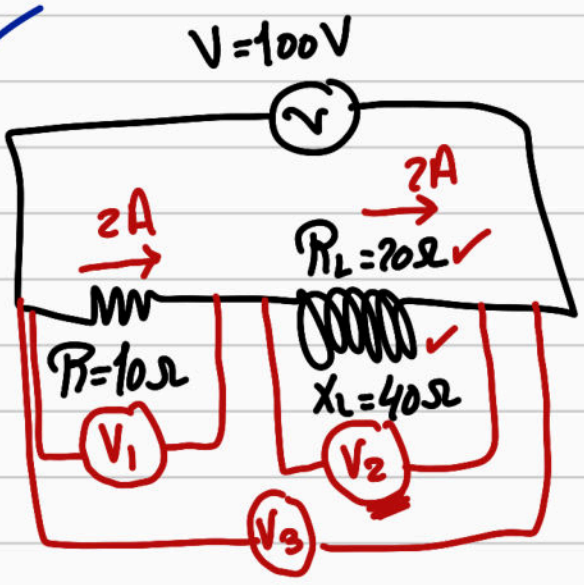
example 7 $V_3 = V_t = 100V$ ✓

Find $V_1 / V_2 / V_3 / P_w / \theta$



$\tan\theta = \frac{X_L}{R}$ (X_L → V_L, R → V_R)

$\tan\theta = \frac{4\phi}{3\phi}$



$Z = \sqrt{R^2 + X_L^2} = 50\Omega$

$I = \frac{100}{50} = 2A$

$P_w = I^2 R = 2^2 \times 30 = 120W$

$2^2 \times 10 + 2^2 \times 20 = 120W$

$V_1 = 2 \times 10 = 20V$

Physics $V_2 = 2 \times \sqrt{20^2 + 40^2} = 40\sqrt{5}\,V$

$V_2 = \sqrt{40^2 + 80^2} = 40\sqrt{5}\,V$

Math $V_2 = \sqrt{100^2 - 20^2} = 40\sqrt{6}\,V$

$V_L = 80V$

$V_{coil} = \sqrt{40^2 + 80^2} = 40\sqrt{5}\,V$

$V_t = \sqrt{60^2 + 80^2} = 100V$ → Source

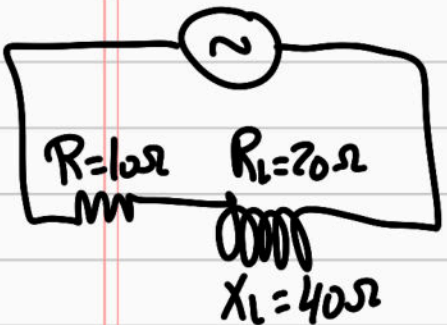
$V_{R_L} = 40V$

$V_R = 20V$

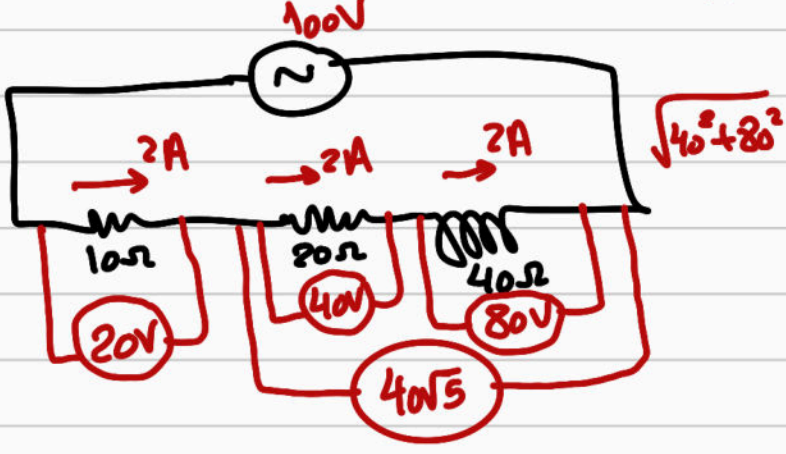
$V_R = 20 + 40 = 60V$

$I = 2A$





→



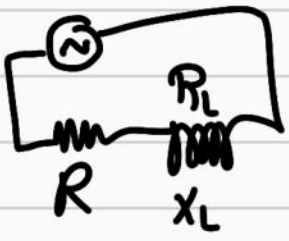
$$I = \frac{V_R}{R} = \frac{V_L}{X_L} = \frac{V_t}{Z}$$

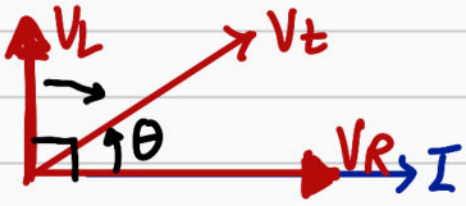
$$X_L = 2\pi f L \qquad X_C = \frac{1}{2\pi f C}$$

RL-Circuit





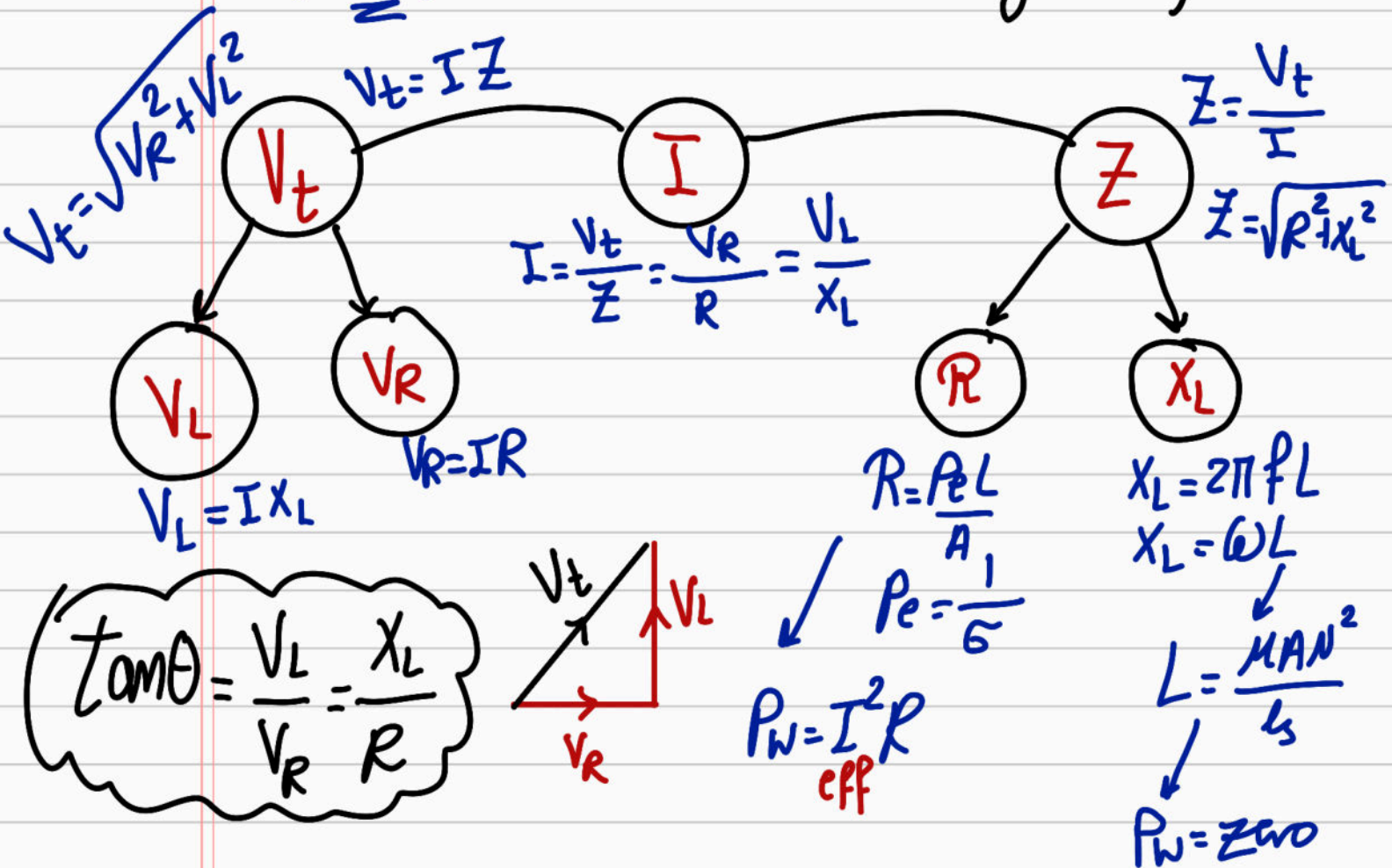


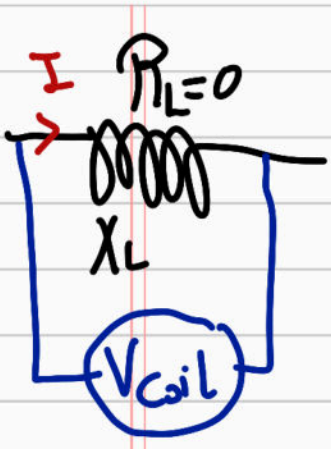


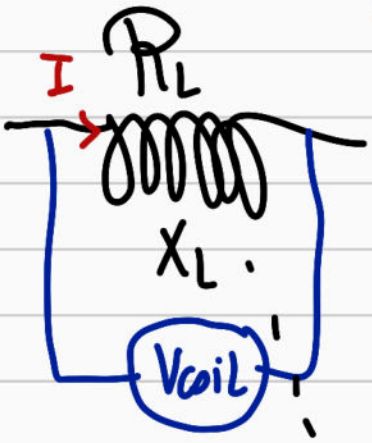
V_t Lead V_R or I by θ

V_t Lag V_L by $90-\theta$

Mind Map





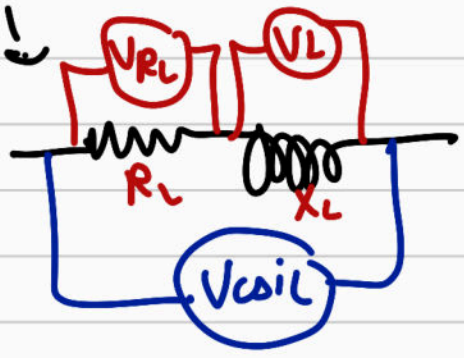


$$V_{coil} = I\,Z_{coil}$$

$$V_{coil} = \sqrt{V_{R_L}^2 + V_{X_L}^2}$$

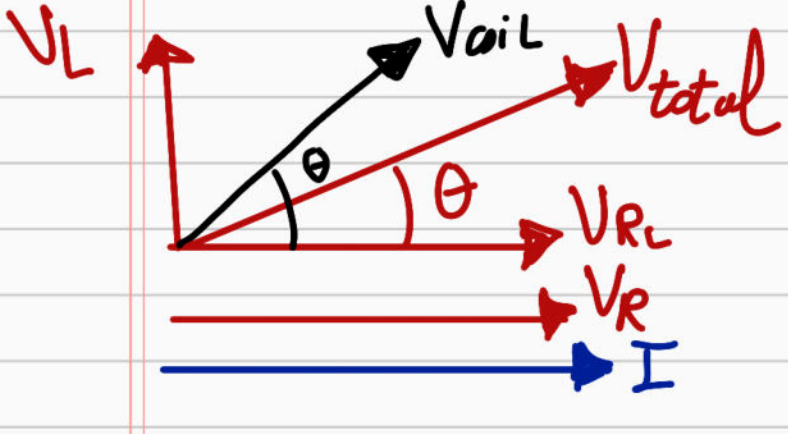
$$V_{coil} = \sqrt{R^2 + X_L^2}$$

$$V_{coil} = V_L = I\,X_L$$



$$V_{total} = I\,Z_{circuit}$$

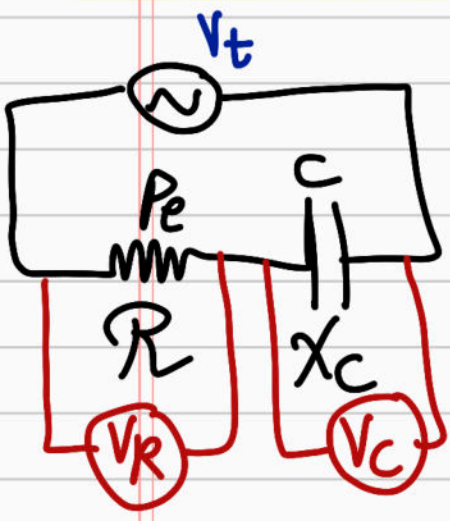
$$V_{coil} = I\,Z_{coil}$$

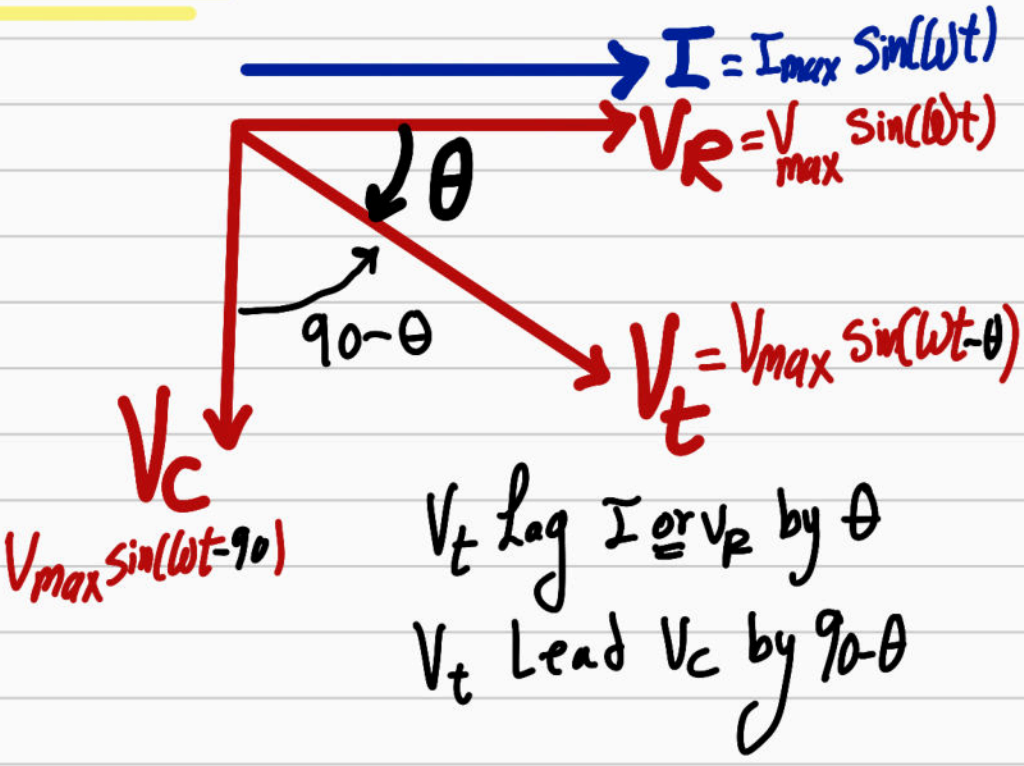


$$\tan\theta = \frac{X_L}{R_L}$$

$$\tan\theta = \frac{X_L}{R + R_L}$$

2 RC – Circuit:



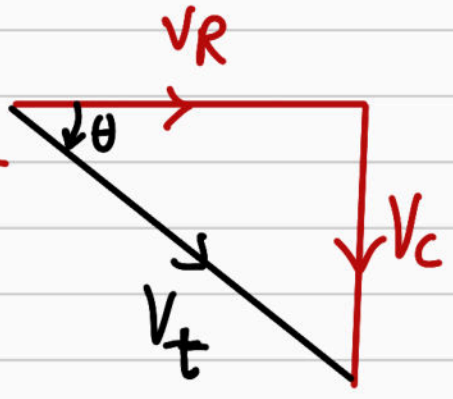


V_t lag I or V_R by θ

V_t Lead V_C by $90-\theta$

#Rules:

1. $V_R = IR$
2. $V_C = IX_C$
3. $V_t = IZ$
4. $V_t = \sqrt{V_R^2 + V_C^2}$
5. $P_w = I_{eff}^2 R$
6. $Z = \sqrt{R^2 + X_C^2}$



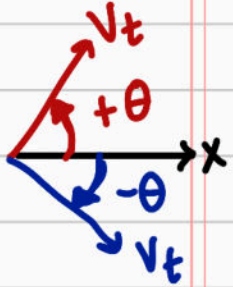
lag ← $\tan\theta = \frac{-V_C = -IX_C}{V_R} = \frac{-X_C}{R}$ → Lag

$\tan\theta = \frac{-X_C}{R}$ → $X_C = \sqrt{3}$, $R = 1\Omega$

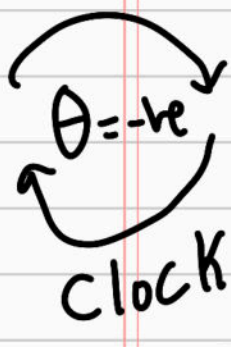
300° 120°

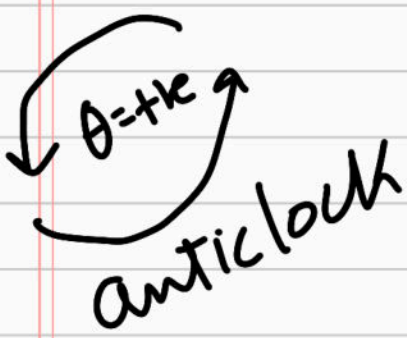
$\tan(x) = -\sqrt{3}$

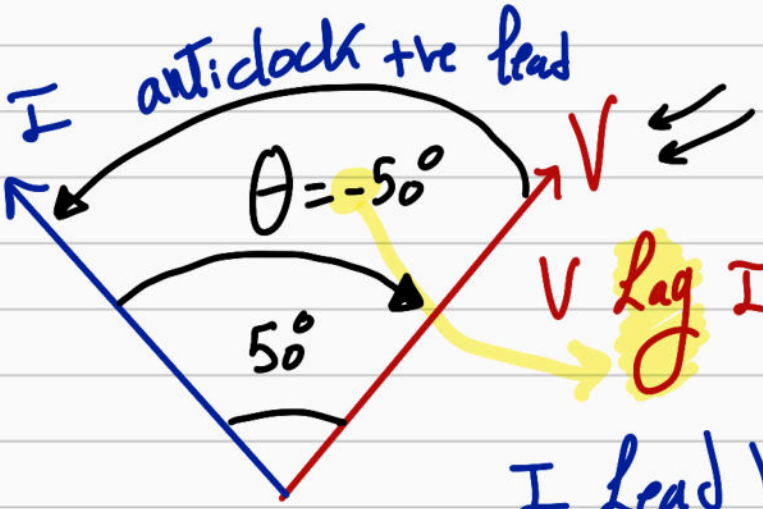
$-60°$



$\tan\theta = \frac{-X_C}{R} \rightarrow \theta = -60°$

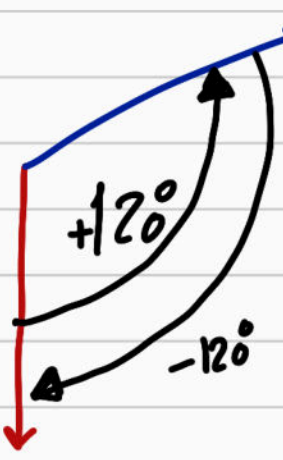






V lag I by 50°

I lead V by 50°

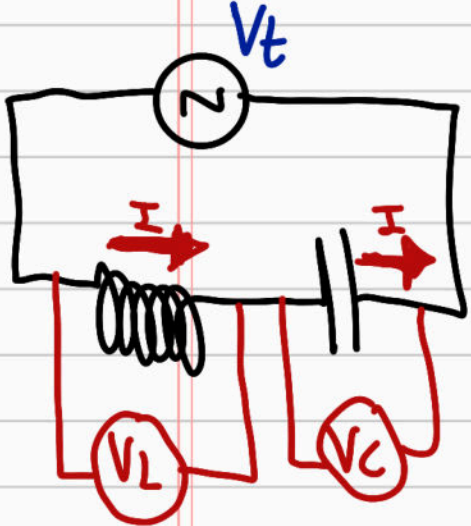


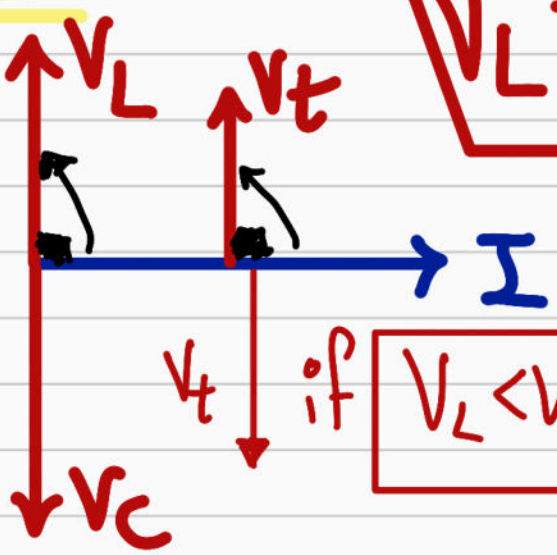
I lead V by 120°

V lag I by 120°

---

③ LC-Circuit:





if $V_L > V_C$ → Vt lead I by 90°

if $V_L < V_C$ → Vt lag I by 90°

① $V_L = I X_L$

② $V_C = I X_C$

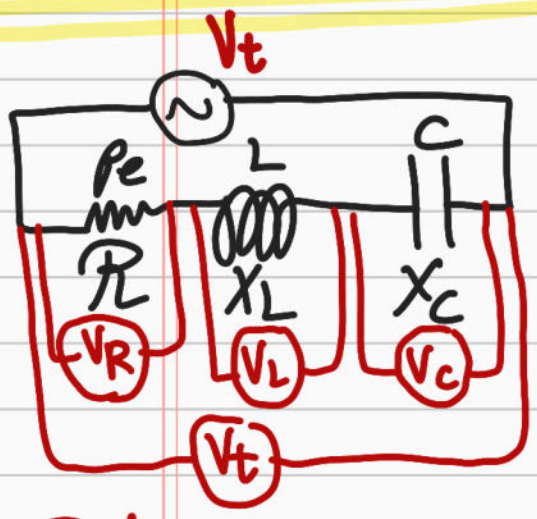
③ $V_t = V_L - V_C$ or $V_C - V_L$

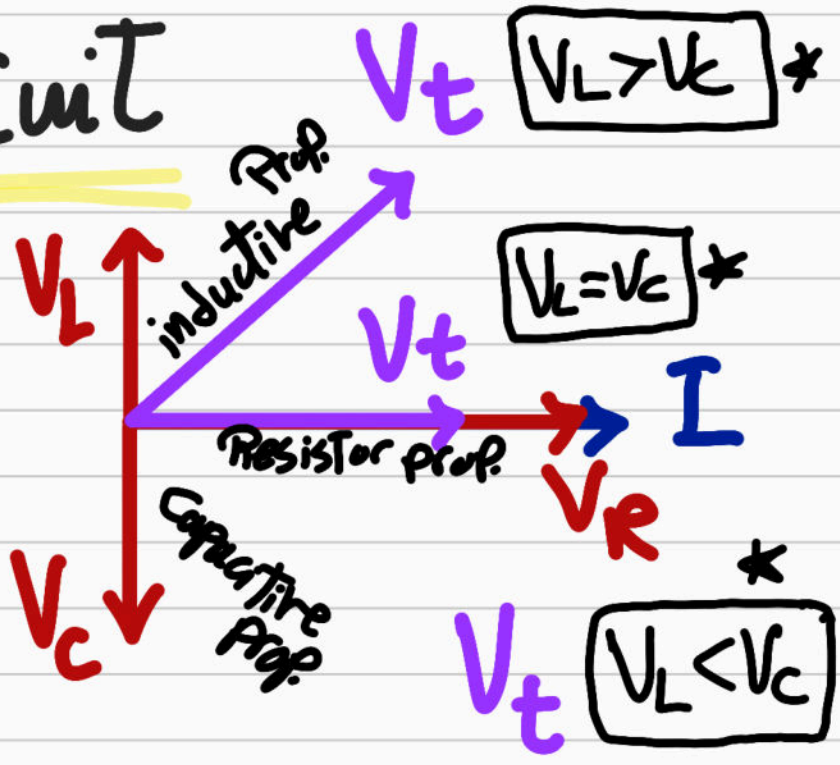
④ $Z = X_L - X_C$ or $X_C - X_L$

⑤ P_W = Zero

⑥ $V_t = IZ$

RLC-Circuit





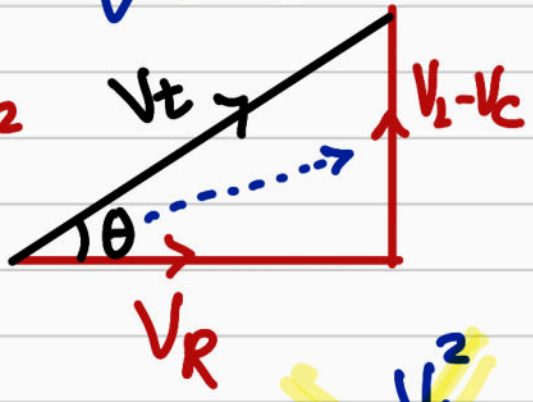
#Rules:

① $V_R = IR$, $R = \frac{\rho_e L}{A}$, $\rho_e = \frac{1}{\sigma}$, $P_w = I_{eff}^2 R$

② $V_L = IX_L$, $X_L = 2\pi f L$, $L = \frac{\mu A N^2}{l_s}$, $P_w = zero$

③ $V_C = IX_C$, $X_C = \frac{1}{2\pi f C}$, $C = \frac{Q}{V}$, $P_w = zero$

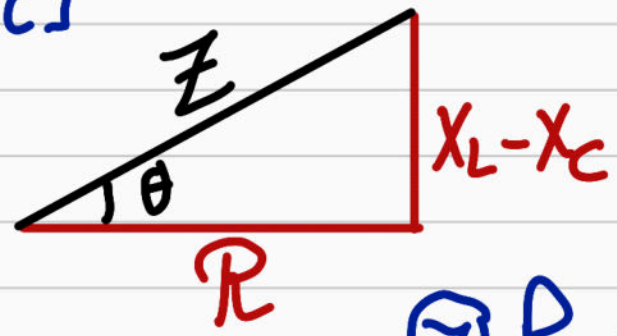
④ $V_t = IZ = \sqrt{V_R^2 + [V_L - V_C]^2}$



⑤ $Z = \sqrt{R^2 + [X_L - X_C]^2}$

⑥ $\tan\theta = \frac{V_L - V_C}{V_R}$

$\tan\theta = \frac{X_L - X_C}{R}$



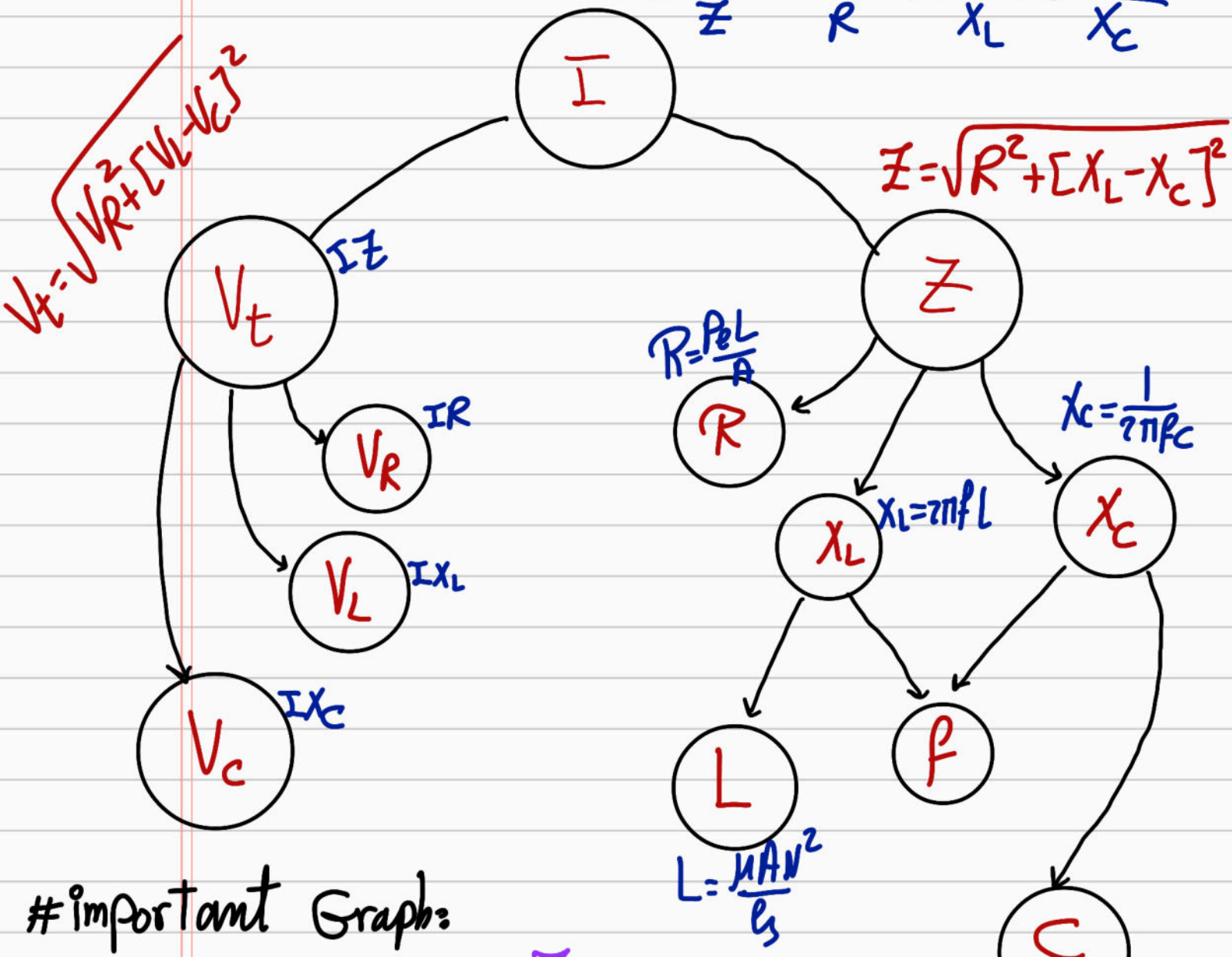
~~$P_w = \frac{V_t^2}{R}$~~

⑦ $P_w = I_{eff}^2 R$

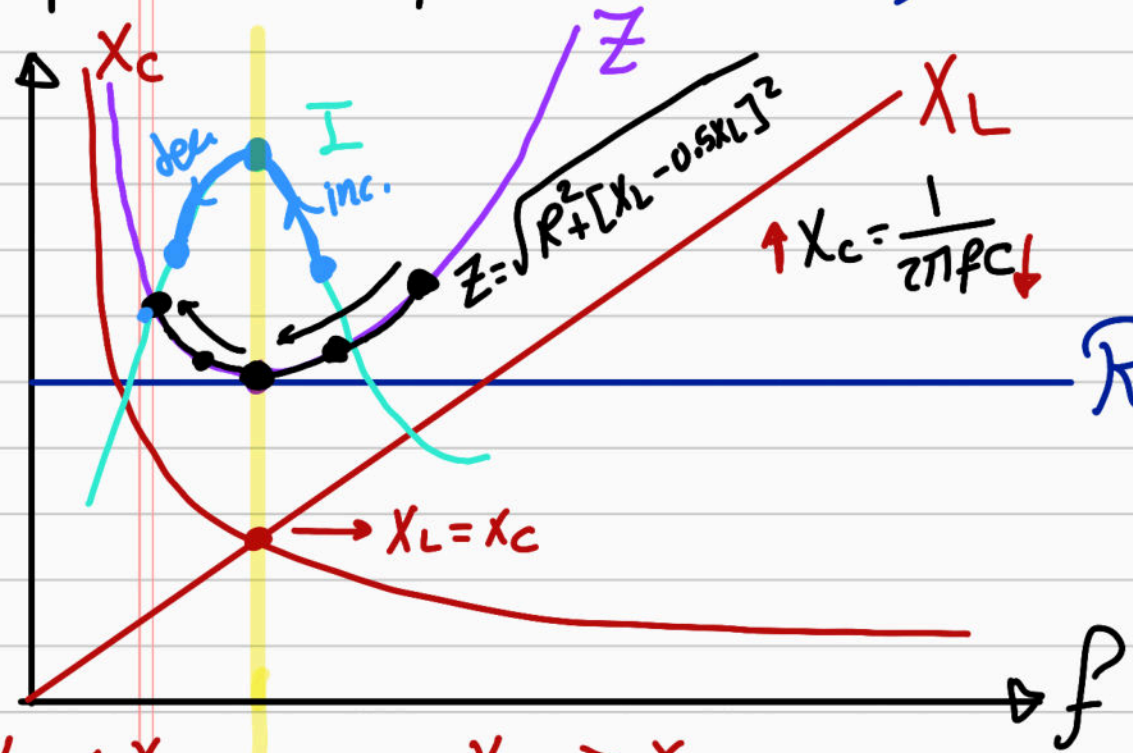
$$I = \frac{V_t}{Z} = \frac{V_R}{R} = \frac{V_L}{X_L} = \frac{V_C}{X_C}$$

$$V_t = \sqrt{V_R^2 + [V_L - V_C]^2}$$

$$Z = \sqrt{R^2 + [X_L - X_C]^2}$$



#important Graph:



$X_L < X_C$

$X_C = 1.5 X_L$

$X_L > X_C$

$X_C = \frac{1}{2} X_L$

$$Z=\sqrt{1^2+[1-0.5]^2}=1.1 \qquad I=\frac{10}{1.1}=9.9A$$

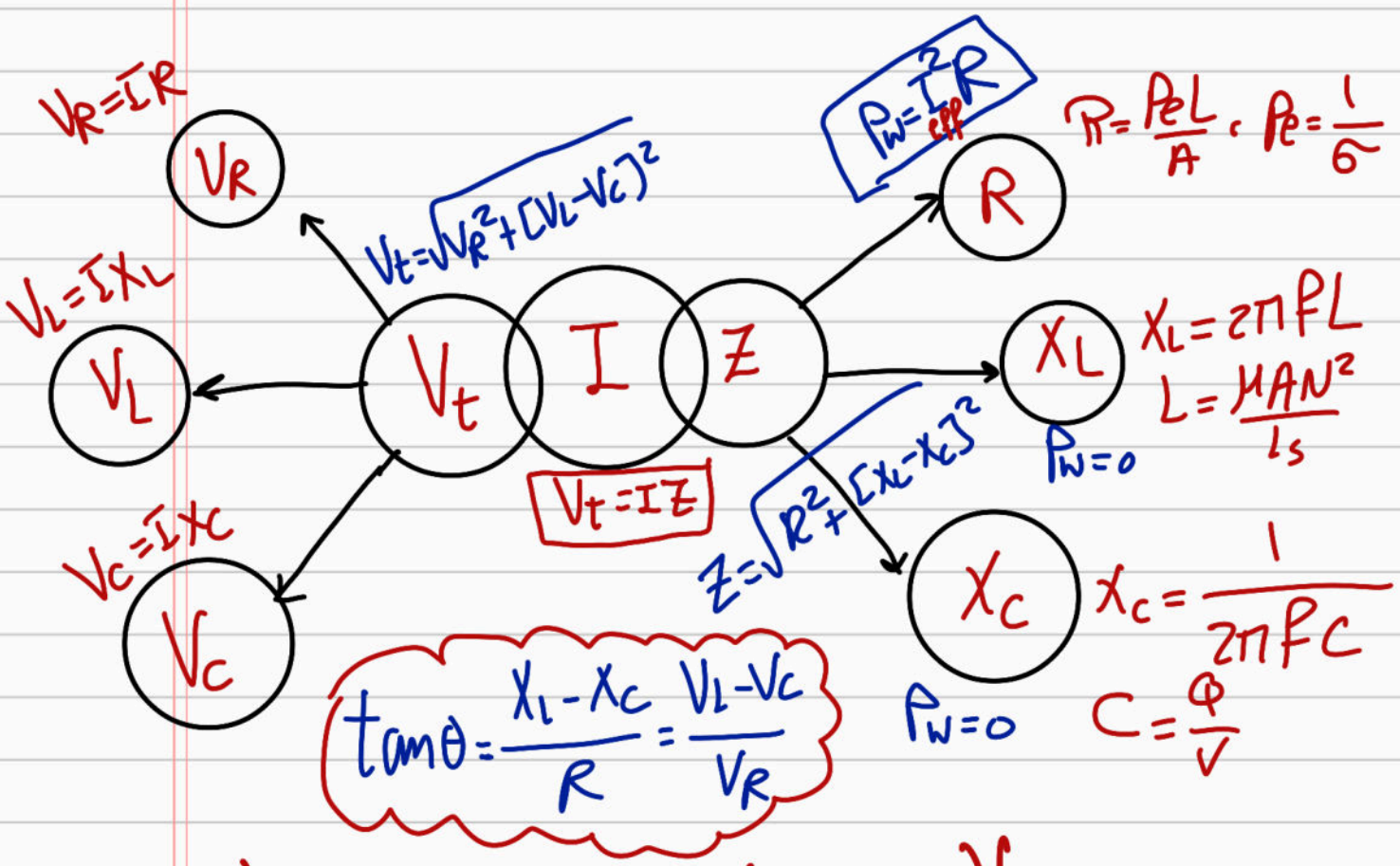
(R, X_L, X_C)

$\uparrow X_C=\frac{1}{2\pi fC}\downarrow$ dec. inc.

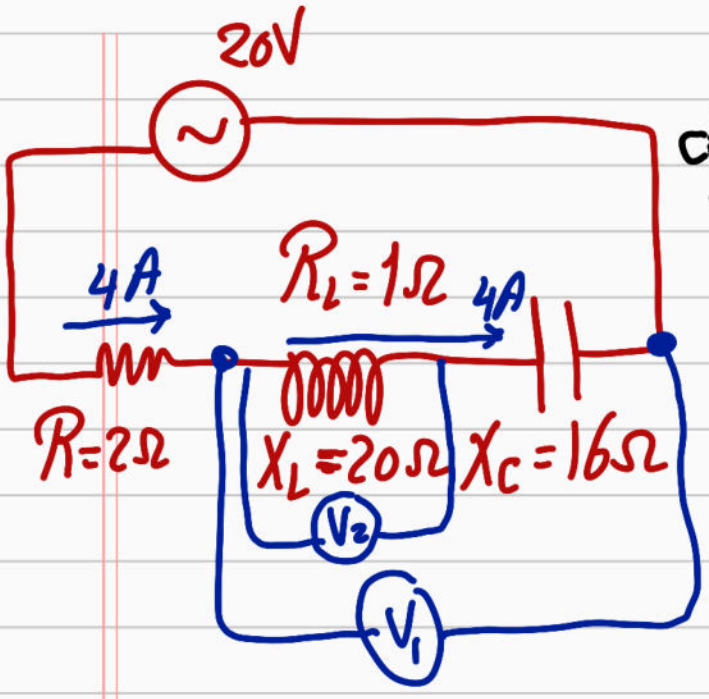
$$Z=\sqrt{1^2+[1-1]^2}=1 \qquad I=\frac{10}{1}=10A$$

inc. dec.

$$Z=\sqrt{1^2+[1-1.5]^2}=1.1 \qquad I=\frac{10}{1.1}=9.9A$$



$$I=\frac{V_t}{Z}=\frac{V_R}{R}=\frac{V_L}{X_L}=\frac{V_C}{X_C}$$

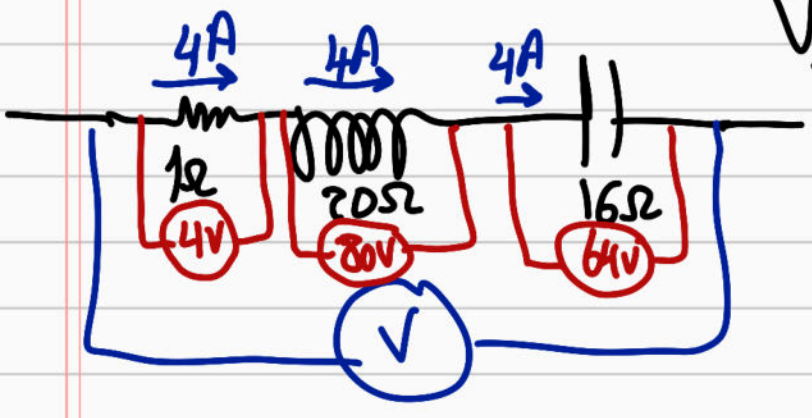


$$Z_{\text{Circuit}} = \sqrt{[1+2]^2 + [20-16]^2} = 5\Omega$$

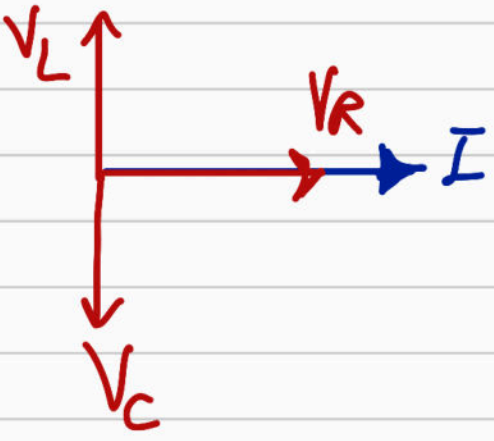
$$I = \frac{V_T}{Z} = \frac{20}{5} = 4A$$

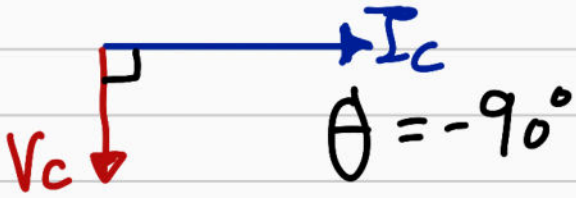
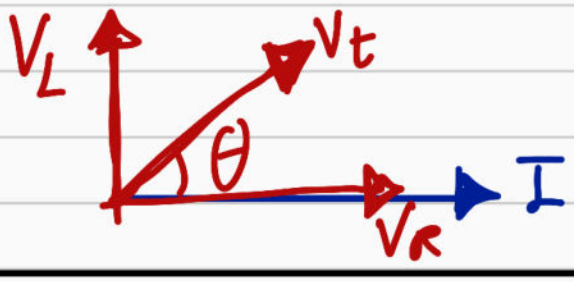
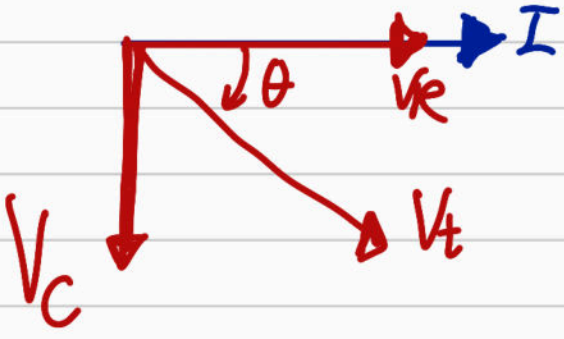
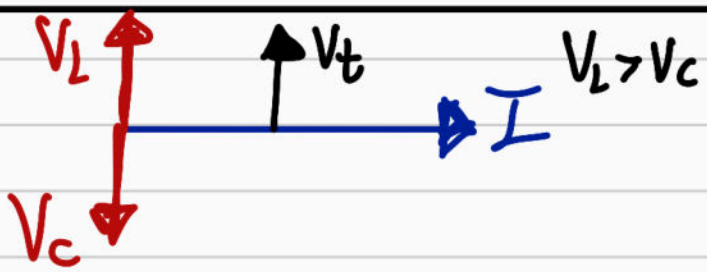
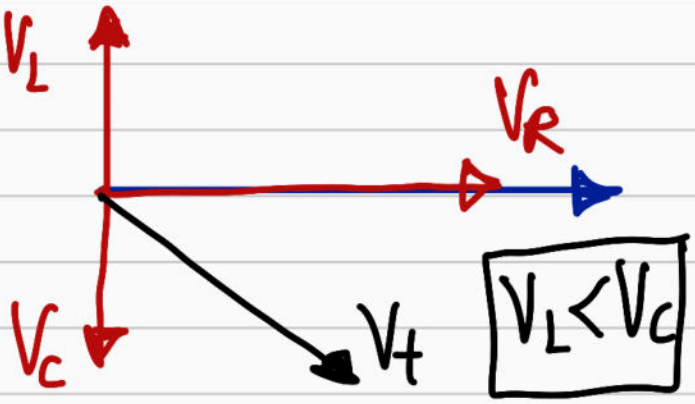
$$V_1 = I Z_{\text{Voltmeter}}$$

$$= 4 \times \sqrt{1^2 + [20-16]^2}$$

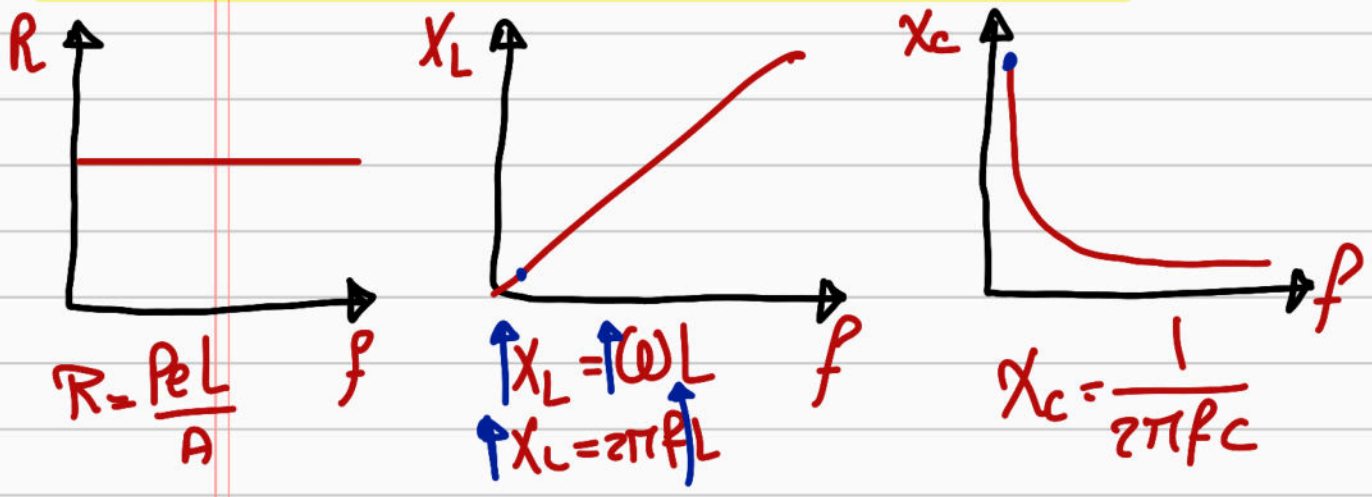


$$V_1 = \sqrt{4^2 + [80-64]^2}$$



Circuit	Phasor diagram	NOTES
R		V_R & I in same Phase $\theta = 0$
L		V_L Lead I by 90° I Lag V_L by 90°
C		V_C Lag I by 90° I lead V_C by 90°
RL		V_t lead I by θ I lag V_t by θ
RC		V_t lag I by θ I lead V_t by θ
LC		$V_L > V_C \rightarrow \theta = +90^\circ$ $V_L < V_C \rightarrow \theta = -90^\circ$
RLC		$V_L > V_C \rightarrow \theta > 0^\circ$ $V_L = V_C \rightarrow \theta = 0^\circ$ $V_L < V_C \rightarrow \theta < 0^\circ$

Chapter (4) Lesson (3) "Resonance"



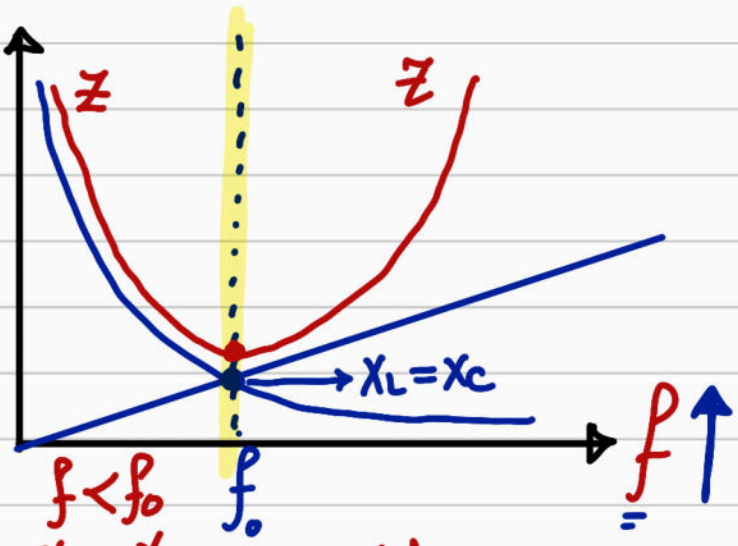
$$Z = \sqrt{R^2 + [X_L - X_C]^2}$$

at low frequencies $X_L < X_C$

$$Z = \sqrt{R^2 + [X_C - X_L]^2}$$

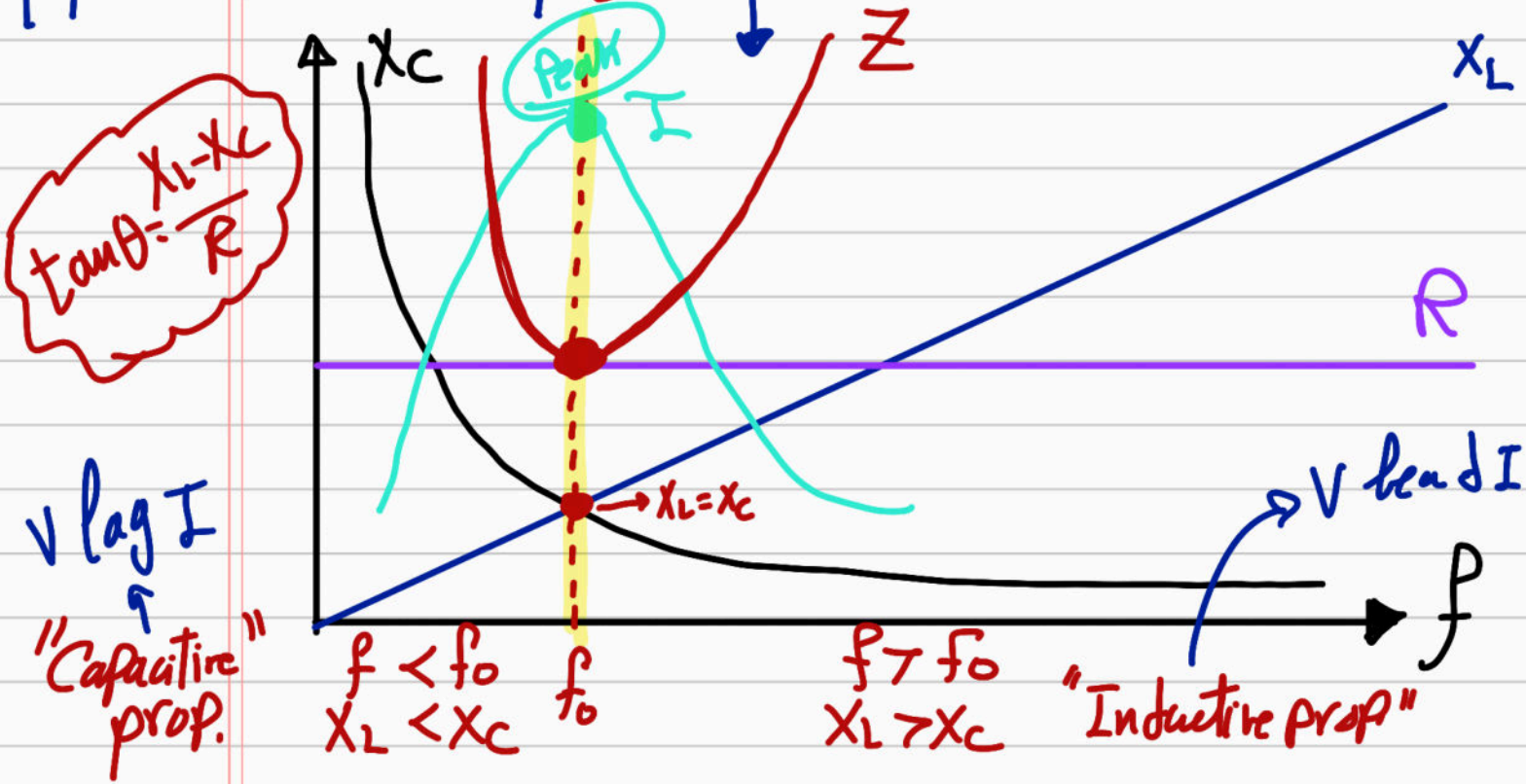
at high frequencies $X_L > X_C$

$$Z = \sqrt{R^2 + [X_L - X_C]^2}$$

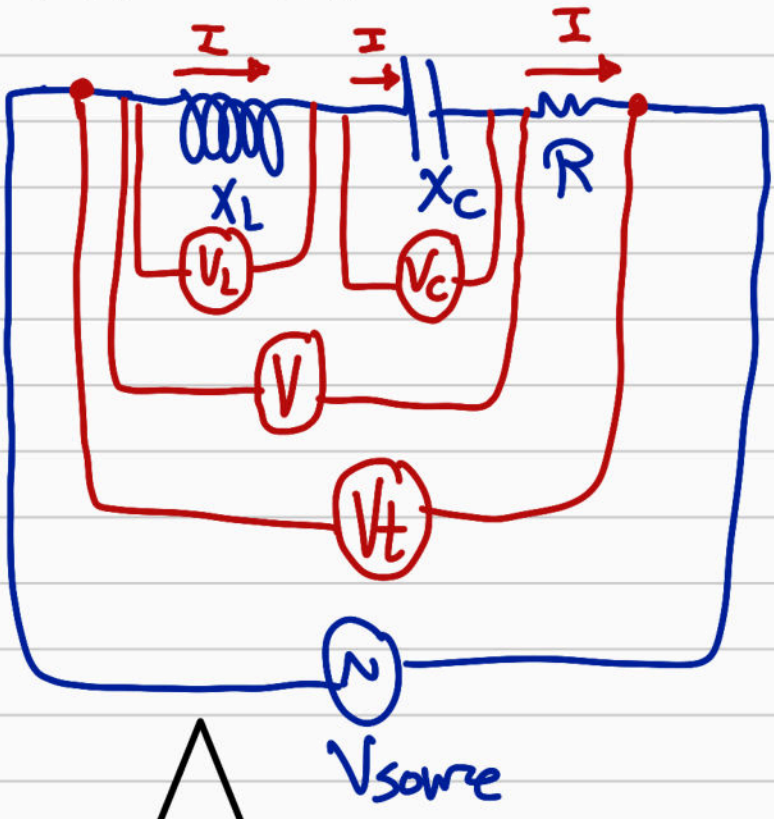


$V_t \rightarrow$ Source

$I = \frac{V_t}{Z}$



#Notes for Solving Resonance:



① $X_L = X_C$

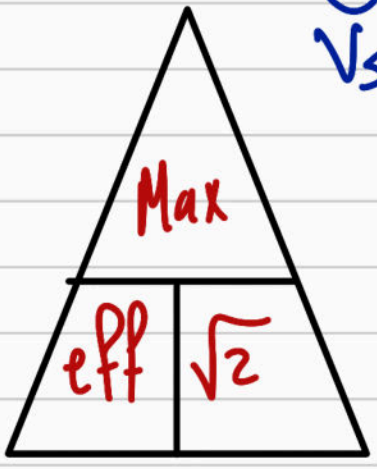
② $V_L = V_C$

③ $V = V_L - V_C = \text{Zero}$

④ $V_t = V_R = V_{source}$

⑤ $Z_{min} = R$

⑥ $I_{eff} = \frac{V_{source}}{Z_{min}}$ (I_{eff}: Peak; Z_{min})



⑦ Power consumed = $I_{eff}^2 R$ (Power consumed: Peak; I_{eff}: Peak)

⑧ Circuit ohmic Resistance properties

a) $\theta_{phase} = \text{Zero}$ b) V_t & V_R & I in same phase

⑨ $f_{Source\ [send]} = f_{Circuit\ [recive]}$

⑩ $f_o = \frac{1}{2\pi\sqrt{LC}}$

$$X_L = X_C$$

$$\omega L = \frac{1}{\omega C}$$

$$\omega^2 LC = 1$$

$$\omega^2 = \frac{1}{LC}$$

$$(2\pi f)^2 = \frac{1}{LC}$$

$$2\pi f = \frac{1}{\sqrt{LC}}$$

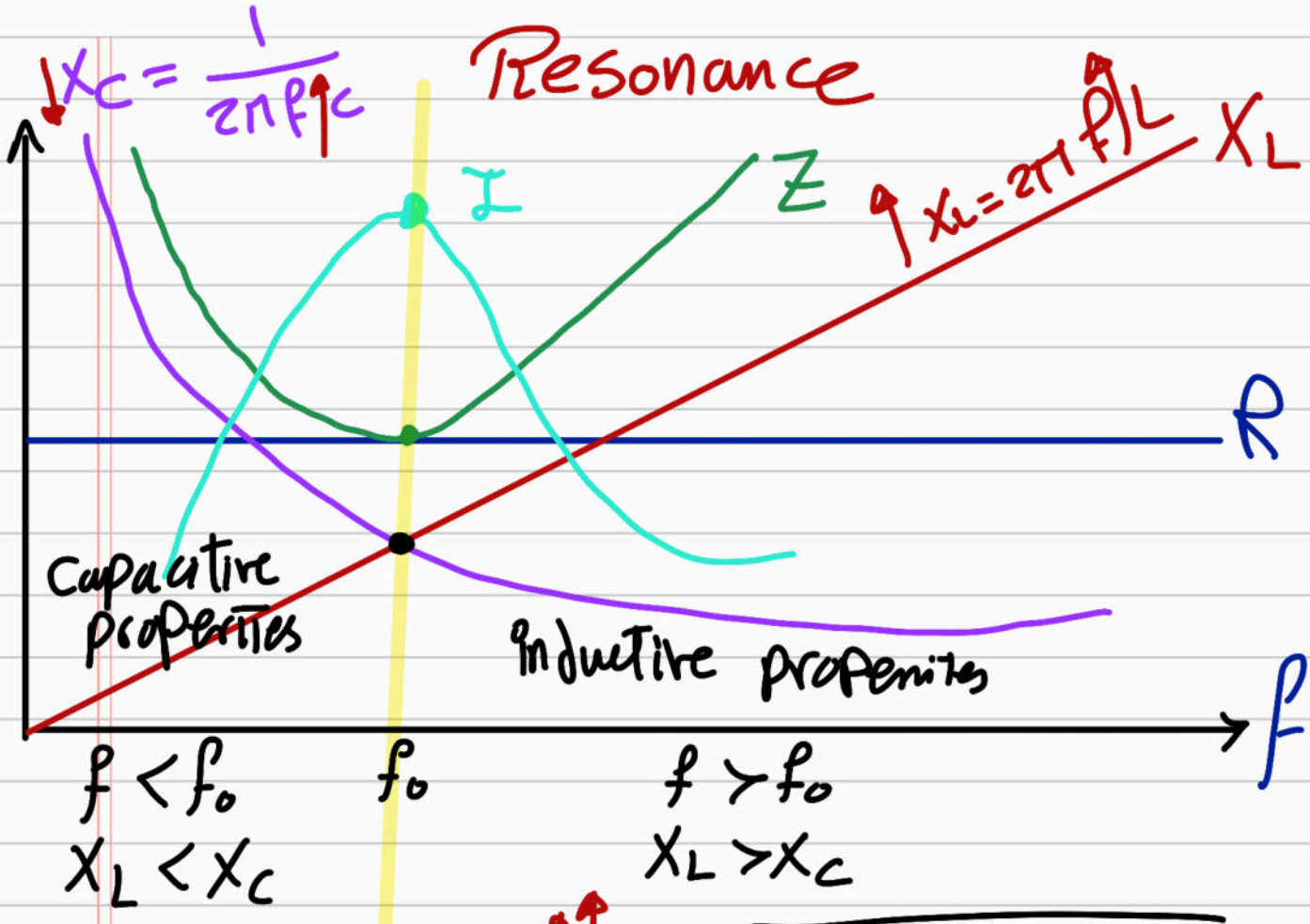
$$f = \frac{1}{2\pi\sqrt{LC}}$$

$$2\pi f L = \frac{1}{2\pi f C}$$

$$4\pi^2 f^2 LC = 1$$

$$f^2 = \frac{1}{4\pi^2 LC}$$

$$f = \frac{1}{2\pi\sqrt{LC}}$$



$$Z = \sqrt{R^2 + [X_C - X_L]^2}$$

$$Z = \sqrt{R^2 + [X_L - X_C]^2}$$

#Notes

Radio

$f_{send} = f_{receive}$

$f_o = \frac{1}{2\pi\sqrt{LC}}$

$V_t = \sqrt{V_R^2 + [V_L - V_C]^2}$

$V_L = V_C$

$V_t = V_R = V_{source}$

$\div I$

$Z = \sqrt{R^2 + [X_L - X_C]^2}$

$X_L = X_C$

$Z_{min} = R$

ohmic Resistance properties

V_t & I in phase

$\theta_{phase} = 0$

$R_L = 0$

$V = V_L - V_C = 0$ ✓

$I = \frac{V_t}{Z}$

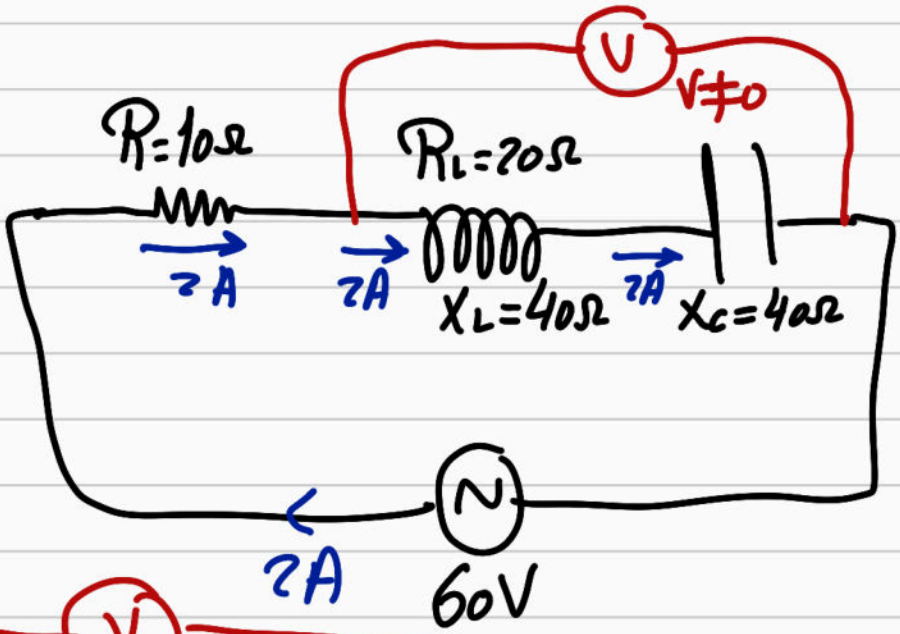
$I_{eff\ Peak} = \frac{V_t}{R}$

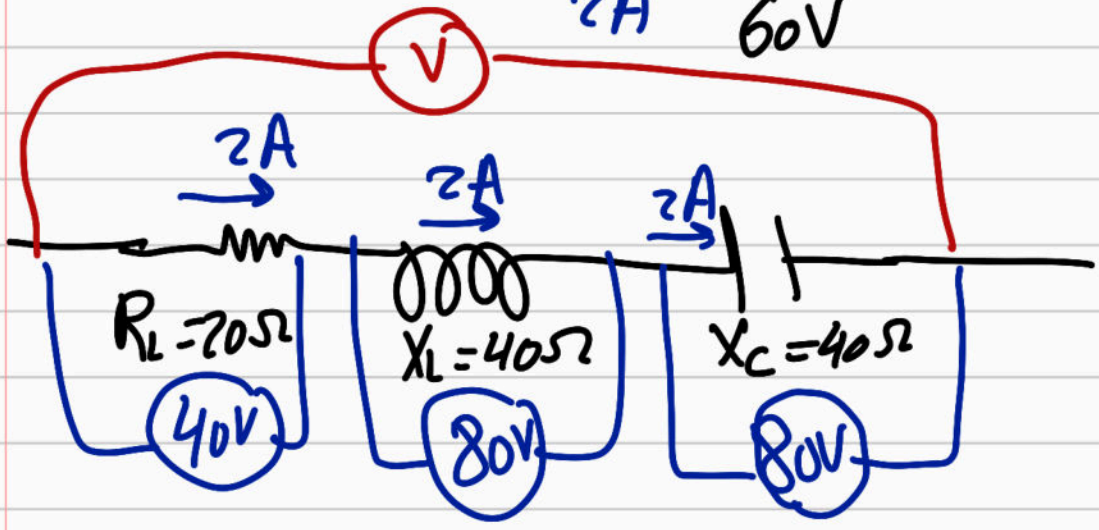
$P_{W\ Peak} = I^2_{eff\ peak} R$

Resonance

$X_L = X_C = 40\Omega$

$I = \frac{V_t}{R} = \frac{60}{30} = 2A$

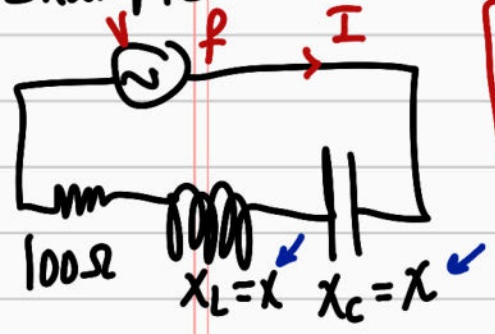




$$V_t = \sqrt{V_R^2 + [V_L - V_C]^2}$$

$$= \sqrt{40^2 + [80-80]^2} = 40V$$

Example



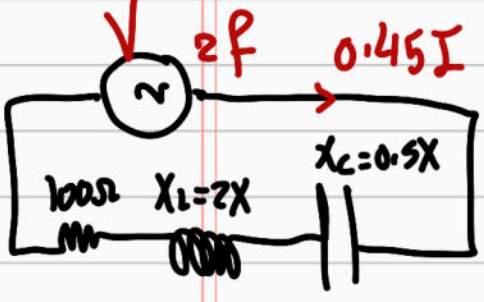
Resonance

$$I = \frac{V}{100} \quad (1)$$

(1) ÷ (2)

$$\frac{1}{0.45} = \frac{\sqrt{100^2 + [1.5X]^2}}{100}$$

$X = 132\Omega$ → X_L, X_C



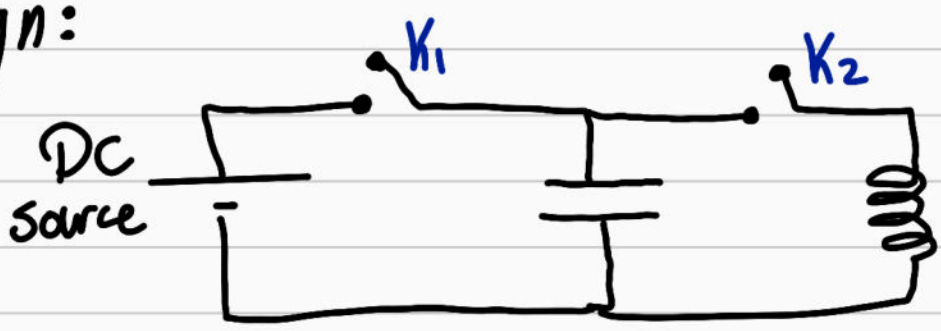
$$0.45I = \frac{V}{\sqrt{100^2 + [2X - 0.5X]^2}} \quad (2)$$

#oscillating circuit:

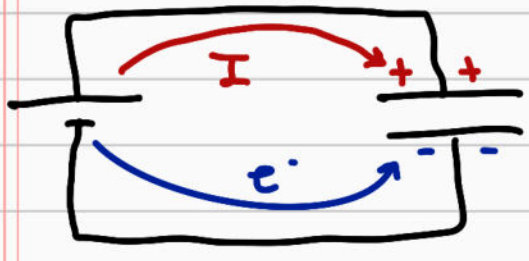
#Sci. idea: Resonance + Energy conversion Elec. ⇌ Magnetic

use: Signal Transmitting (دائرة إرسال)

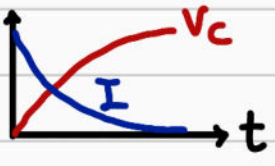
design:



K1 closed, K2 open

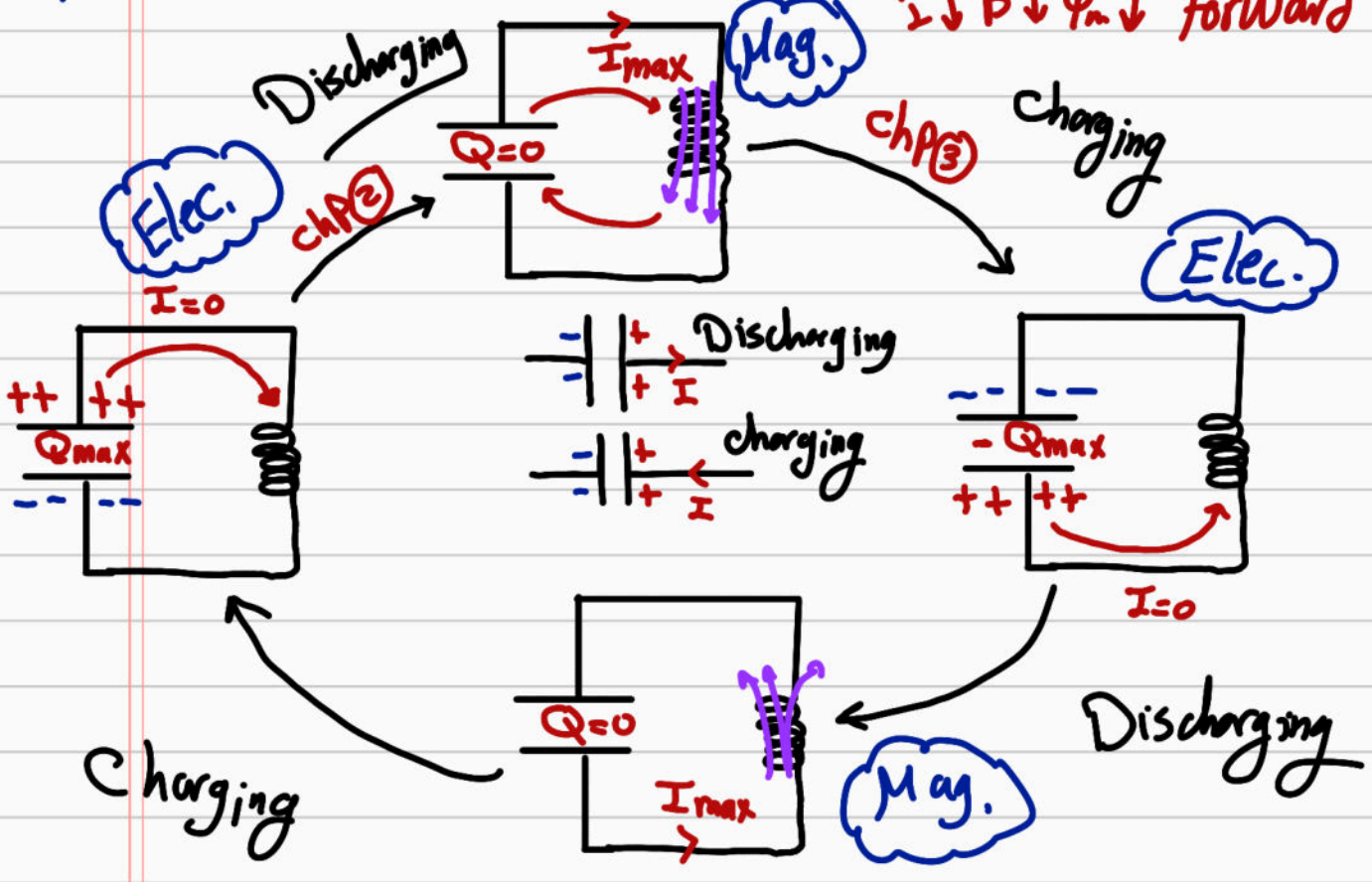


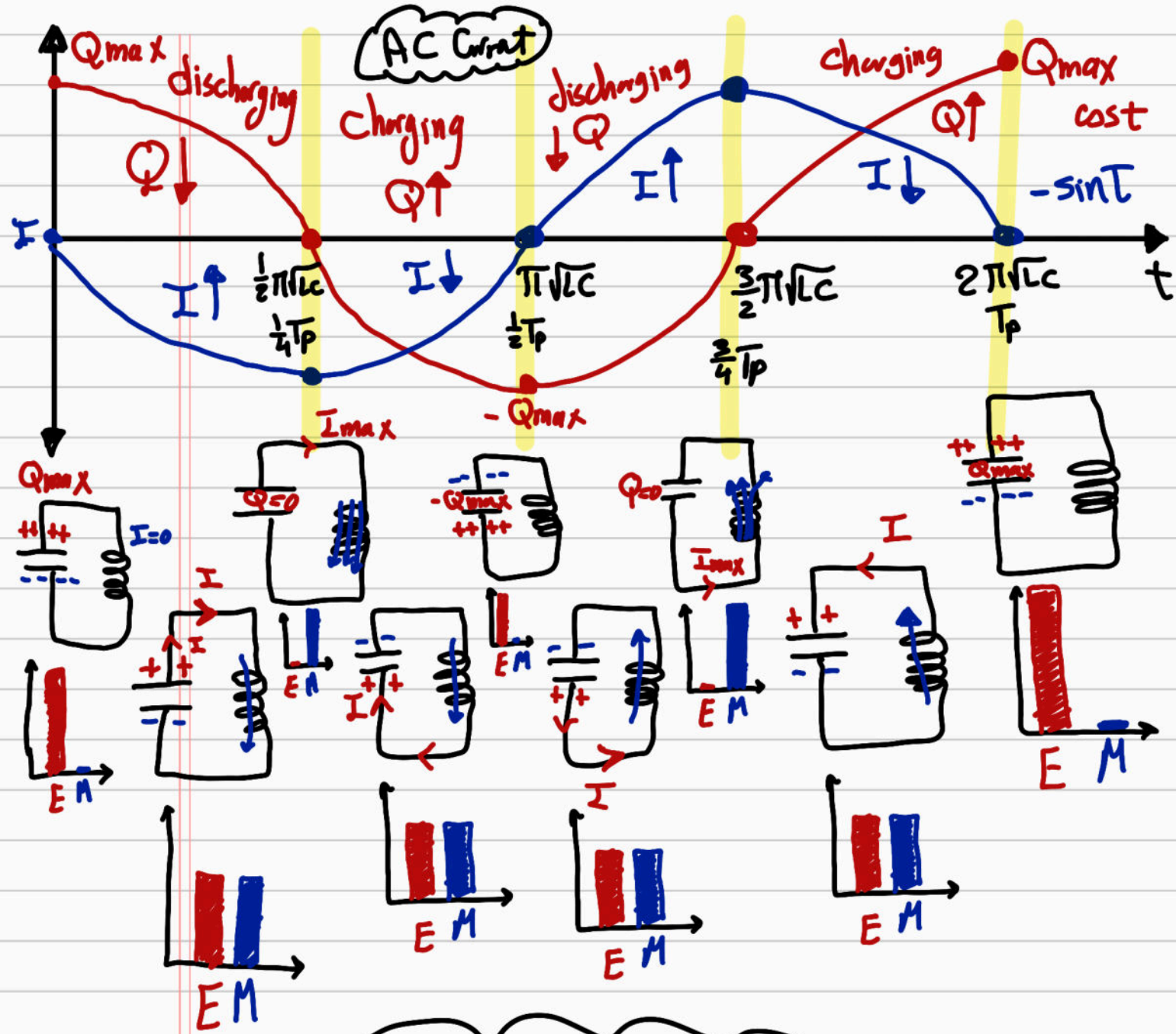
Capacitor is charging



K1 open, K2 closed

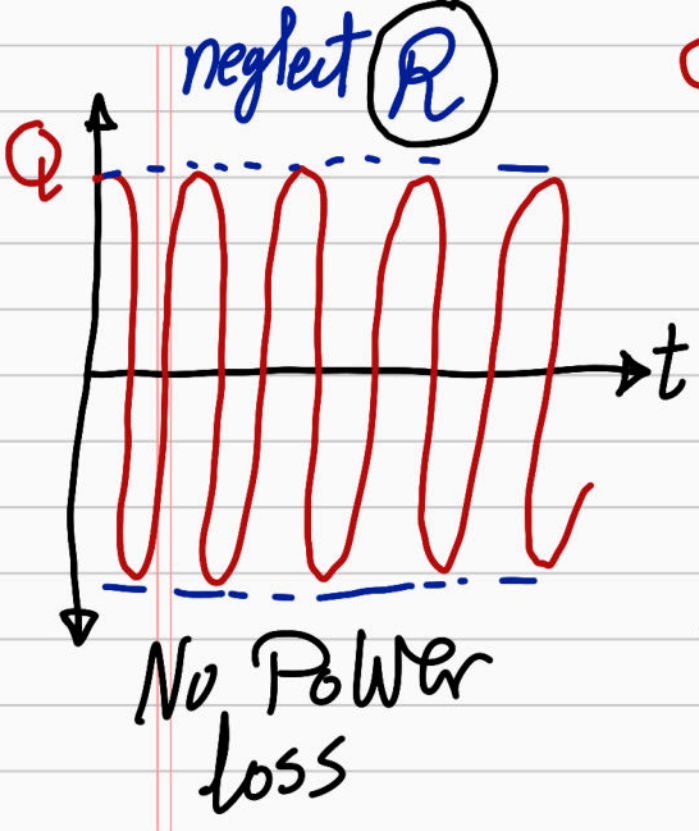
I↓ B↓ ϕ_m↓ forward

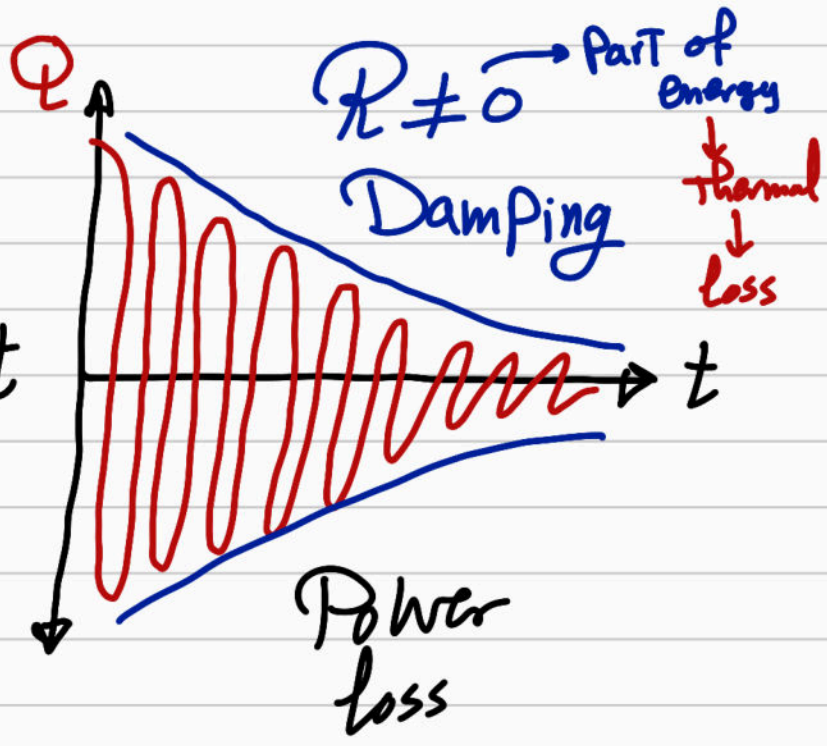




$$f = \frac{1}{2\pi\sqrt{LC}}$$

$$T_p = 2\pi\sqrt{LC}$$





#Radio / Tuning Circuit:

